Team notebook

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1 Data Structure

1.1 Fenwick2D

```
struct Fenwick2D {
#define gb(x) (x) & -(x)
  vector<vector<int>> nodes;
  vector<vector<int>> bit;
  int sx;
  void init(int _sx) {
    sx = _sx;
    nodes.resize(sx + 1);
    bit.resize(sx + 1);
  }
  void init_nodes() {
    for (int i = 1; i <= sx; ++i) {
      sort(all(nodes[i]));
      nodes[i].resize(unique(all(nodes[i])) -
           nodes[i].begin());
      bit[i].resize(sz(nodes[i]) + 1);
    }
  }
  void fake_update(int x, int y) {
    for (; x <= sx; x += gb(x)) nodes[x].push_back(y);
  }
  void fake_get(int x, int y) {
    for (; x > 0; x -= gb(x)) nodes[x].push_back(y);
  }
  void update(int x, int yy, int val) {
    for (; x <= sx; x += gb(x))
      for (int y = lower_bound(all(nodes[x]), yy) -
           nodes[x].begin() + 1;
          y <= sz(nodes[x]); y += gb(y))
        bit[x][y] = max(bit[x][y], val);
  }
  int get(int x, int yy) {
    int res = 0;
    for (; x > 0; x -= gb(x))
      for (int y = upper_bound(all(nodes[x]), yy) -
           nodes[x].begin(); y > 0;
          y -= gb(y))
        res = max(res, bit[x][y]);
    return res;
  }
};
```

1.2 ImplicitTreap

```
// Implicit Treap
// Tested: https://oj.vnoi.info/problem/sqrt_b
struct Treap {
  ll val;
  int prior, size;
  ll sum;
  Treap *left, *right;
  Treap(ll val)
      : val(val), prior(rng()), size(1), sum(val),
           left(NULL), right(NULL){};
};
int size(Treap *t) { return t == NULL ? 0 : t->size; }
void down(Treap *t) {
  // do lazy propagation here
}
void refine(Treap *t) {
  if (t == NULL) return;

  t->size = 1;
  t->sum = t->val;
  if (t->left != NULL) {
    t->size += t->left->size;
    t->sum += t->left->sum;
  }
  if (t->right != NULL) {
    t->size += t->right->size;
    t->sum += t->right->sum;
  }
}
void split(Treap *t, Treap *&left, Treap *&right, int val)
     {
  if (t == NULL) return void(left = right = NULL);
  down(t);
  if (size(t->left) < val) {
    split(t->right, t->right, right, val - size(t->left) -
          1);
    left = t;
  } else {
    split(t->left, left, t->left, val);
    right = t;
  }
  refine(t);
}
void merge(Treap *&t, Treap *left, Treap *right) {
  if (left == NULL) {
    t = right;
    return;
  }
  if (right == NULL) {
    t = left;
    return;
  }

  down(left);
  down(right);
  if (left->prior < right->prior) {
    merge(left->right, left->right, right);
    t = left;
  } else {
    merge(right->left, left, right->left);
    t = right;
  }
  refine(t);
}
array<Treap *, 2> split(Treap *root, int val) {
  array<Treap *, 2> t;
  split(root, t[0], t[1], val);
  return t;
}
array<Treap *, 3> split(Treap *root, int l, int r) {
  array<Treap *, 3> t;
  Treap *tmp;
  split(root, t[0], t[1], l - 1);
  tmp = t[1];
  split(tmp, t[1], t[2], r - l + 1);

  return t;
}
Treap *root;
```

1.3 MeldableHeap

```
mt19937 gen(0x94949);
template<typename T>
struct Node {
        Node *l, *r;
        T v;
        Node(T x): l(0), r(0), v(x){}
};
template<typename T>
Node<T>* Meld(Node<T>* A, Node<T>* B) {
        if(!A) return B; if(!B) return A;
        if(B->v < A->v) swap(A, B);
        if(gen()&1) A->l = Meld(A->l, B);
        else A->r = Meld(A->r, B);
        return A;
}
template<typename T>
struct Heap {
        Node<T> *r; int s;
        Heap(): r(0), s(0){}
        void push(T x) {
                r = Meld(new Node<T>(x), r);
                ++s;
        }
        int size(){ return s; }
        bool empty(){ return s == 0;}
        T top(){ return r->v; }
        void pop() {
                Node<T>* p = r;
                r = Meld(r->l, r->r);
                delete p;
                --s;
        }
        void Meld(Heap x) {
                s += x->s;
                r = Meld(r, x->r);
        }
};
```

1.4 OrderStatisticTree

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template <class T>
using Tree =
```

```
    tree<T, null_type, less<T>, rb_tree_tag,
         tree_order_statistics_node_update>;
void example() {
  Tree<int> t, t2;
  t.insert(8);
  auto it = t.insert(10).first;
  assert(it == t.lower_bound(9));
  assert(t.order_of_key(10) == 1);
  assert(t.order_of_key(11) == 2);
  assert(*t.find_by_order(0) == 8);
  t.join(t2); // assuming T < T2 or T > T2, merge t2 into t
}
```

1.5 PalindromeTree

```
template <int MAXC = 26>
struct PalindromicTree {
  PalindromicTree(const string& str) : _sz(str.size() +
       5), next(_sz, vector<int>(MAXC, 0)), link(_sz, 0),
       qlink(_sz, 0), cnt(_sz, 0), right_id(_sz, 0),
       len(_sz, 0), s(_sz, 0) {
    init();
    for (int i = 0; i < (int)str.size(); ++i) {
      add(str[i], i);
    }
    count();
  }
  int _sz;

  // returns vector of (left, right, frequency)
  vector<tuple<int, int, int>> get_palindromes() {
    vector<tuple<int, int, int>> res;
    dfs(0, res);
    dfs(1, res);
    return res;
  }

  void dfs(int u, vector<tuple<int, int, int>>& res) {
    if (u > 1) { // u = 0 and u = 1 are two empty nodes
      res.emplace_back(right_id[u] - len[u] + 1,
           right_id[u], cnt[u]);
    }
    for (int i = 0; i < MAXC; ++i) {
      if (next[u][i]) dfs(next[u][i], res);
    }
  }

  int last, n, p;
  vector<vector<int>> next, dlink;
  vector<int> link, qlink, cnt, right_id, len, s;

  int newnode(int l, int right) {
    len[p] = l;
    right_id[p] = right;
    return p++;
  }
  void init() {
    p = 0;
    newnode(0, -1), newnode(-1, -1);
    n = last = 0;
    s[n] = -1, link[0] = 1;
  }
  int getlink(int x) {
    while (s[n - len[x] - 1] != s[n]) {
      if (s[n - len[link[x]] - 1] == s[n])
        x = link[x];
      else
        x = qlink[x];
    }
    return x;
  }
  void add(char c, int right) {
    c -= 'a';
    s[++n] = c;
    int cur = getlink(last);
    if (!next[cur][(int)c]) {
      int now = newnode(len[cur] + 2, right);
      link[now] = next[getlink(link[cur])][(int)c];
      next[cur][(int)c] = now;
      if (s[n - len[link[now]]] == s[n -
           len[link[link[now]]]]) {
        qlink[now] = qlink[link[now]];
      } else {
        qlink[now] = link[link[now]];
      }
    }
    last = next[cur][(int)c];
    cnt[last]++;
  }
  void count() {
    for (int i = p - 1; i >= 0; i--) {
      cnt[link[i]] += cnt[i];
    }
  }
};
```

1.6 RMQ

```
// RMQ O(1): 1-indexed
// remember to change the constants, types
using ll = long long;
#define For(i, j, k) for ( int i = (j) ; i <= (k) ; i++ )
#define Fol(i, j, k) for ( int i = (j) ; i >= (k) ; i-- )
namespace RMQ
{
       using T = int; constexpr int N = 2e6 + 6; // change
            this
       inline bool cmp(T x, T y) { return x < y; } //
            change to '>' to query max
       inline T calc(T x, T y) { return cmp(x, y) ? x : y;
            }
       T val[N], pre[N], st[__lg(( N >> 5 ) + 9) + 1][( N
            >> 5 ) + 9]; unsigned f[N];
       __attribute__((target("bmi" ))) inline int
            lb(unsigned x) { return __builtin_ctz(x) ; }
       __attribute__((target("lzcnt"))) inline int
            hb(unsigned x) { return __builtin_clz(x) ^ 31;
            }
       inline void build(int n, T *a)
       {
              int m = ( n - 1 ) >> 5, o = hb(m + 1),
                   stk[33]; copy(a + 1, a + n + 1, val);
              For(i, 0, n - 1) pre[i] = i & 31 ?
                   calc(pre[i - 1], val[i]) : val[i];
              For(i, 0, m) st[0][i] = pre[min(n - 1, i <<
                   5 | 31)];
              For(i, 1, o) For(j, 0, m + 1 - ( 1 << i ))
                     st[i][j] = calc(st[i - 1][j], st[i -
                          1][j + ( 1 << ( i - 1 ) )]);
              For(i, 0, n - 1)
                     if ( i & 31 )
                     {
                            f[i] = f[i - 1];
                            while ( o &&
                                 !cmp(val[stk[o]],
                                 val[i]) ) f[i] &= ~( 1u
                                 << ( stk[o--] & 31 ) );
                            f[i] |= 1u << ( ( stk[++o] =
                                 i ) & 31 );
                     }
                     else f[i] = 1u << ( ( stk[o = 1] = i
                          ) & 31 );
       }
       inline T qry(int l, int r)
       {
              if ( ( --l >> 5 ) == ( --r >> 5 ) ) return
                   val[l + lb(f[r] >> ( l & 31 ))];
              T z = calc(pre[r], val[l + lb(f[l | 31] >> (
                   l & 31 ))]);
              if ( ( l = ( l >> 5 ) + 1 ) == ( r >>= 5 ) )
                   return z;
              int t = hb(r - l); return calc(z,
                   calc(st[t][l], st[t][r - ( 1 << t )]));
       }
}
// build: RMQ::build(n, a), a is an array (not a vector!)
// query: RMQ::qry(l, r)
```

1.7 SegmentTree

```
struct Tree {
  typedef int T;
  static constexpr T unit = INT_MIN;
  T f(T a, T b) { return max(a, b); } // (any associative
       fn)
  vector<T> s;
  int n;
  Tree(int n = 0, T def = unit) : s(2 * n, def), n(n) {}
  void update(int pos, T val) {
    for (s[pos += n] = val; pos /= 2;) s[pos] = f(s[pos *
         2], s[pos * 2 + 1]);
  }
  T query(int b, int e) { // query [b , e)
    T ra = unit, rb = unit;
    for (b += n, e += n; b < e; b /= 2, e /= 2) {
      if (b % 2) ra = f(ra, s[b++]);
      if (e % 2) rb = f(s[--e], rb);
```

```
    }
    return f(ra, rb);
  }
};
```

1.8 WalkOnBIT

```
template<class T>
class FenwickTree {
public:
	int n;
	vector<T> bit;
	int LOG;
	FenwickTree(int _n) {
		init(_n);
	}
	void init(int _n) {
		n = _n;
		bit = vector<T>(n, 0);
		LOG = __lg(n);
	}
	T sum(int idx) {
		assert(0 <= idx && idx < n);
		T ret = 0;
		for (; idx >= 0; idx &= idx + 1, --idx)
			ret += bit[idx];
		return ret;
	}
	T sum(int l, int r) {
		return sum(r) - sum(l - 1);
	}
	void add(int idx, int delta) {
		for (; idx < n; idx |= idx + 1)
			bit[idx] += delta;
	}
	int bit_search(T v) { // lower_bound in the prefix
	    sums array of A
		T sum = 0;
		int pos = -1;
		for(int i = LOG; i >= 0; i--){
			if (pos + (1 << i) < n && sum +
			    bit[pos + (1 << i)] < v) {
				sum += bit[pos + (1 << i)];
				pos += (1 << i);
			}
		}
		return pos + 1; // +1 because 'pos' will
		    have position of largest value less
		    than 'v'
	}
};
```

2 DP

2.1 DivideAndConquerDP

```
const ll INF = 1e18;

void calc(int i, int l, int r, int optL, int optR) {
  if (l > r) return;
  int mid = (l + r) / 2;
  f[i][mid] = INF; // change to -INF to find max
  int opt = -1;
  for (int k = optL; k <= min(mid, optR); ++k) {
    ll c = f[i - 1][k] + cost(k + 1, mid);
    if (c < f[i][mid]) {
      f[i][mid] = c;
      opt = k;
    }
  }
  calc(i, l, mid - 1, optL, opt);
  calc(i, mid + 1, r, opt, optR);
}
```

2.2 LineContainer

```
struct Line {
	mutable ll k, m, p;
	bool operator<(const Line& o) const { return k <
	    o.k; } // change '<' to '>' to get min
	bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
	// (for doubles, use inf = 1/.0, div(a,b) = a/b)
	static const ll inf = LLONG_MAX;
	ll div(ll a, ll b) { // floored division
		return a / b - ((a ^ b) < 0 && a % b); }
	bool isect(iterator x, iterator y) {
		if (y == end()) return x->p = inf, 0;
		if (x->k == y->k) x->p = x->m > y->m ? inf :
		    -inf;
		else x->p = div(y->m - x->m, x->k - y->k);
		return x->p >= y->p;
	}
	void add(ll k, ll m) {
		auto z = insert({k, m, 0}), y = z++, x = y;
		while (isect(y, z)) z = erase(z);
		if (x != begin() && isect(--x, y)) isect(x,
		    y = erase(y));
		while ((y = x) != begin() && (--x)->p >=
		    y->p)
			isect(x, erase(y));
	}
	ll query(ll x) {
		assert(!empty());
		auto l = *lower_bound(x);
		return l.k * x + l.m;
	}
};
```

2.3 lis

```
template<typename T>
int lis(const vector<T>& a) {
  vector<T> u;
  for (const T& x : a) {
    auto it = lower_bound(u.begin(), u.end(), x);
    if (it == u.end()) {
      u.push_back(x);
    } else {
      *it = x;
    }
  }
  return (int) u.size();
}
```

3 Geometry

3.1 AngleBisector

```
// bisector vector of <abc
PT angle_bisector(PT &a, PT &b, PT &c){
  PT p = a - b, q = c - b;
  return p + q * sqrt(dot(p, p) / dot(q, q));
}
```

3.2 Centroid

```
// centroid of a (possibly non-convex) polygon,
// assuming that the coordinates are listed in a clockwise
    or
// counterclockwise fashion. Note that the centroid is
    often known as
// the "center of gravity" or "center of mass".
PT centroid(vector<PT> &p) {
    int n = p.size(); PT c(0, 0);
    double sum = 0;
    for (int i = 0; i < n; i++) sum += cross(p[i], p[(i +
        1) % n]);
    double scale = 3.0 * sum;
    for (int i = 0; i < n; i++) {
        int j = (i + 1) % n;
        c = c + (p[i] + p[j]) * cross(p[i], p[j]);
    }
    return c / scale;
}
```

3.3 ClosestPair

```
typedef Point<ll> P;
pair<P, P> closest(vector<P> v) {
  assert(sz(v) > 1);
  set<P> S;
```

```
  sort(all(v), [](P a, P b) { return a.y < b.y; });
  pair<ll, pair<P, P>> ret{LLONG_MAX, {P(), P()}};
  int j = 0;
  for (P p : v) {
    P d{1 + (ll)sqrt(ret.first), 0};
    while (v[j].y <= p.y - d.x) S.erase(v[j++]);
    auto lo = S.lower_bound(p - d), hi = S.upper_bound(p +
        d);
    for (; lo != hi; ++lo) ret = min(ret, {(*lo -
        p).dist2(), {*lo, p}});
    S.insert(p);
  }
  return ret.second;
```

3.4 ConvexPolygon

```
// checks if convex or not
bool is_convex(vector<PT>& p) {
  bool s[3];
  s[0] = s[1] = s[2] = 0;
  int n = p.size();
  for (int i = 0; i < n; i++) {
    int j = (i + 1) % n;
    int k = (j + 1) % n;
    s[sign(cross(p[j] - p[i], p[k] - p[i])) + 1] = 1;
    if (s[0] && s[2])
      return 0;
  }
  return 1;
}
// -1 if strictly inside, 0 if on the polygon, 1 if
    strictly outside
// it must be strictly convex, otherwise make it strictly
    convex first
int is_point_in_convex(vector<PT>& p, const PT& x) { //
    O(log n)
  int n = p.size();
  assert(n >= 3);
  int a = orientation(p[0], p[1], x), b =
      orientation(p[0], p[n - 1], x);
  if (a < 0 || b > 0)
    return 1;
  int l = 1, r = n - 1;
  while (l + 1 < r) {
    int mid = l + r >> 1;
    if (orientation(p[0], p[mid], x) >= 0) l = mid;
    else r = mid;
  }
  int k = orientation(p[l], p[r], x);
  if (k <= 0)
    return -k;
  if (l == 1 && a == 0)
    return 0;
  if (r == n - 1 && b == 0)
    return 0;
  return -1;
}
```

3.5 ExtremeVertex

```
// id of the vertex having maximum dot product with z
// polygon must need to be convex
// top - upper right vertex
// for minimum dot prouct negate z and return -dot(z,
    p[id])
int extreme_vertex(vector<PT> &p, const PT &z, const int
    top) { // O(log n)
  int n = p.size();
  if (n == 1) return 0;
  double ans = dot(p[0], z); int id = 0;
  if (dot(p[top], z) > ans) ans = dot(p[top], z), id =
      top;
  int l = 1, r = top - 1;
  while (l < r) {
    int mid = l + r >> 1;
    if (dot(p[mid + 1], z) >= dot(p[mid], z)) l = mid +
        1;
    else r = mid;
  }
  if (dot(p[l], z) > ans) ans = dot(p[l], z), id = l;
  l = top + 1, r = n - 1;
  while (l < r) {
    int mid = l + r >> 1;
    if (dot(p[(mid + 1) % n], z) >= dot(p[mid], z)) l =
        mid + 1;
    else r = mid;
  }
  l %= n;
  if (dot(p[l], z) > ans) ans = dot(p[l], z), id = l;
  return id;
}
```

3.6 GeometricMedian

```
// it returns a point such that the sum of distances
// from that point to all points in p is minimum
// O(n log^2 MX)
PT geometric_median(vector<PT> p) {
  auto tot_dist = [&](PT z) {
    double res = 0;
    for (int i = 0; i < p.size(); i++) res +=
        dist(p[i], z);
    return res;
  };
  auto findY = [&](double x) {
    double yl = -1e5, yr = 1e5;
    for (int i = 0; i < 60; i++) {
      double ym1 = yl + (yr - yl) / 3;
      double ym2 = yr - (yr - yl) / 3;
      double d1 = tot_dist(PT(x, ym1));
      double d2 = tot_dist(PT(x, ym2));
      if (d1 < d2) yr = ym2;
      else yl = ym1;
    }
    return pair<double, double> (yl, tot_dist(PT(x,
        yl)));
  };
  double xl = -1e5, xr = 1e5;
  for (int i = 0; i < 60; i++) {
    double xm1 = xl + (xr - xl) / 3;
    double xm2 = xr - (xr - xl) / 3;
    double y1, d1, y2, d2;
    auto z = findY(xm1); y1 = z.first; d1 = z.second;
    z = findY(xm2); y2 = z.first; d2 = z.second;
    if (d1 < d2) xr = xm2;
    else xl = xm1;
  }
  return {xl, findY(xl).first };
}
```

3.7 GeometryTemplate

```
const long double PI = acos(-1);
struct Vector {
  using type = long long;
  type x, y;
  Vector operator-(const Vector &other) const {
    return {x - other.x, y - other.y};
  }
  type operator*(const Vector &other) const {
    return x * other.y - other.x * y;
  }
  type operator%(const Vector &other) const {
    return x * other.x + y * other.y;
  }
  bool operator==(const Vector &other) const {
    return x == other.x and y == other.y;
  }
  bool operator!=(const Vector &other) const { return
      !(*this == other); }
  friend type cross(const Vector &A, const Vector &B,
      const Vector &C) {
    return (B - A) * (C - A);
  }
  friend type dist(Vector A) { return A.x * A.x + A.y *
      A.y; }
  friend type dot(const Vector &A, const Vector &B, const
      Vector &C) {
    Vector u = (B - A), v = (C - A);
    return u % v;
  }
  friend istream &operator>>(istream &is, Vector &V) {
    is >> V.x >> V.y;
    return is;
  }
  friend ostream &operator<<(ostream &os, Vector &V) {
    os << V.x << ' ' << V.y;
    return os;
  }
  friend double angle(const Vector &A, const Vector &B,
      const Vector &C) {
    double x = dot(B, A, C) / sqrt(dist(A - B) * dist(C -
        B));
    return acos(min(1.0, max(-1.0, x))) * 180.0 / PI;
  }
```

```
};
using Point = Vector;
const Point origin = {0, 0};

long double area(Point A, Point B, Point C) {
  long double res =
      cross(origin, A, B) + cross(origin, B, C) +
          cross(origin, C, A);
  return abs(res) / 2.0;
}
```

3.8 IsPoint

```
// -1 if strictly inside, 0 if on the polygon, 1 if
    strictly outside
int is_point_in_triangle(PT a, PT b, PT c, PT p) {
    if (sign(cross(b - a,c - a)) < 0) swap(b, c);
    int c1 = sign(cross(b - a,p - a));
    int c2 = sign(cross(c - b,p - b));
    int c3 = sign(cross(a - c,p - c));
    if (c1<0 || c2<0 || c3 < 0) return 1;
    if (c1 + c2 + c3 != 3) return 0;
    return -1;
}

bool is_point_on_polygon(vector<PT> &p, const PT& z) {
    int n = p.size();
    for (int i = 0; i < n; i++) {
        if (is_point_on_seg(p[i], p[(i + 1) % n], z))
            return 1;
    }
    return 0;
}

// returns 1e9 if the point is on the polygon
int winding_number(vector<PT> &p, const PT& z) { // O(n)
    if (is_point_on_polygon(p, z)) return 1e9;
    int n = p.size(), ans = 0;
    for (int i = 0; i < n; ++i) {
        int j = (i + 1) % n;
        bool below = p[i].y < z.y;
        if (below != (p[j].y < z.y)) {
            auto orient = orientation(z, p[j], p[i]);
            if (orient == 0) return 0;
            if (below == (orient > 0)) ans += below ? 1 :
                -1;
        }
    }
    return ans;
}
// -1 if strictly inside, 0 if on the polygon, 1 if
    strictly outside
int is_point_in_polygon(vector<PT> &p, const PT& z) { //
    O(n)
    int k = winding_number(p, z);
    return k == 1e9 ? 0 : k == 0 ? 1 : -1;
}
```

3.9 Line

```
struct line {
    PT a, b; // goes through points a and b
    PT v; double c; //line form: direction vec [cross] (x,
        y) = c
    line() {}
    //direction vector v and offset c
    line(PT v, double c) : v(v), c(c) {
        auto p = get_points();
        a = p.first; b = p.second;
    }
    // equation ax + by + c = 0
    line(double _a, double _b, double _c) : v({_b, -_a}),
        c(-_c) {
        auto p = get_points();
        a = p.first; b = p.second;
    }
    // goes through points p and q
    line(PT p, PT q) : v(q - p), c(cross(v, p)), a(p), b(q)
        {}
    pair<PT, PT> get_points() { //extract any two
        points from this line
        PT p, q; double a = -v.y, b = v.x; // ax + by = c
        if (sign(a) == 0) {
            p = PT(0, c / b);
            q = PT(1, c / b);
        }
        else if (sign(b) == 0) {
            p = PT(c / a, 0);
            q = PT(c / a, 1);
        }
        else {
            p = PT(0, c / b);
            q = PT(1, (c - a) / b);
        }
        return {p, q};
        }
    //ax + by + c = 0
    array<double, 3> get_abc() {
        double a = -v.y, b = v.x;
        return {a, b, c};
    }
    // 1 if on the left, -1 if on the right, 0 if on the
        line
    int side(PT p) { return sign(cross(v, p) - c); }
    // line that is perpendicular to this and goes through
        point p
    line perpendicular_through(PT p) { return {p, p +
        perp(v)}; }
    // translate the line by vector t i.e. shifting it by
        vector t
    line translate(PT t) { return {v, c + cross(v, t)}; }
    // compare two points by their orthogonal projection on
        this line
    // a projection point comes before another if it comes
        first according to vector v
    bool cmp_by_projection(PT p, PT q) { return dot(v, p) <
        dot(v, q); }
    line shift_left(double d) {
        PT z = v.perp().truncate(d);
        return line(a + z, b + z);
    }
};
```

3.10 LineLineIntersection

```
// intersection point between ab and cd assuming unique
    intersection exists
bool line_line_intersection(PT a, PT b, PT c, PT d, PT
    &ans) {
    double a1 = a.y - b.y, b1 = b.x - a.x, c1 = cross(a, b);
    double a2 = c.y - d.y, b2 = d.x - c.x, c2 = cross(c, d);
    double det = a1 * b2 - a2 * b1;
    if (det == 0) return 0;
    ans = PT((b1 * c2 - b2 * c1) / det, (c1 * a2 - a1 * c2)
        / det);
    return 1;
}
```

3.11 MaximumCircleCover

```
// find a circle of radius r that contains as many points
    as possible
// O(n^2 log n);
double maximum_circle_cover(vector<PT> p, double r, circle
    &c) {
    int n = p.size();
    int ans = 0;
    int id = 0; double th = 0;
    for (int i = 0; i < n; ++i) {
        // maximum circle cover when the circle goes
            through this point
        vector<pair<double, int>> events = {{-PI, +1}, {PI,
            -1}};
        for (int j = 0; j < n; ++j) {
            if (j == i) continue;
            double d = dist(p[i], p[j]);
            if (d > r * 2) continue;
            double dir = (p[j] - p[i]).arg();
            double ang = acos(d / 2 / r);
            double st = dir - ang, ed = dir + ang;
            if (st > PI) st -= PI * 2;
            if (st <= -PI) st += PI * 2;
            if (ed > PI) ed -= PI * 2;
            if (ed <= -PI) ed += PI * 2;
            events.push_back({st - eps, +1}); // take care
                of precisions!
            events.push_back({ed, -1});
            if (st > ed) {
                events.push_back({-PI, +1});
                events.push_back({+PI, -1});
            }
        }
        sort(events.begin(), events.end());
        int cnt = 0;
        for (auto &&e: events) {
```

```
            cnt += e.second;
            if (cnt > ans) {
                ans = cnt;
                id = i; th = e.first;
            }
        }
    }
    PT w = PT(p[id].x + r * cos(th), p[id].y + r * sin(th));
    c = circle(w, r); //best_circle
    return ans;
}
```

3.12 MaximumInscribedCircle

```
// radius of the maximum inscribed circle in a convex
     polygon
double maximum_inscribed_circle(vector<PT> p) {
    int n = p.size();
    if (n <= 2) return 0;
    double l = 0, r = 20000;
    while (r - l > eps) {
        double mid = (l + r) * 0.5;
        vector<HP> h;
        const int L = 1e9;
        h.push_back(HP(PT(-L, -L), PT(L, -L)));
        h.push_back(HP(PT(L, -L), PT(L, L)));
        h.push_back(HP(PT(L, L), PT(-L, L)));
        h.push_back(HP(PT(-L, L), PT(-L, -L)));
        for (int i = 0; i < n; i++) {
            PT z = (p[(i + 1) % n] - p[i]).perp();
            z = z.truncate(mid);
            PT y = p[i] + z, q = p[(i + 1) % n] + z;
            h.push_back(HP(p[i] + z, p[(i + 1) % n] + z));
        }
        vector<PT> nw = half_plane_intersection(h);
        if (!nw.empty()) l = mid;
        else r = mid;
    }
    return l;
}
```

3.13 MinimumEnclosingCircle

```
// given n points, find the minimum enclosing circle of
     the points
// call convex_hull() before this for faster solution
// expected O(n)
circle minimum_enclosing_circle(vector<PT> &p) {
    random_shuffle(p.begin(), p.end());
    int n = p.size();
    circle c(p[0], 0);
    for (int i = 1; i < n; i++) {
        if (sign(dist(c.p, p[i]) - c.r) > 0) {
            c = circle(p[i], 0);
            for (int j = 0; j < i; j++) {
                if (sign(dist(c.p, p[j]) - c.r) > 0) {
                    c = circle((p[i] + p[j]) / 2, dist(p[i],
                         p[j]) / 2);
                    for (int k = 0; k < j; k++) {
                        if (sign(dist(c.p, p[k]) - c.r) > 0) {
                            c = circle(p[i], p[j], p[k]);
                        }
                    }
                }
            }
        }
    }
    return c;
}
```

3.14 MinimumEnclosingRectangle

```
// minimum perimeter
double minimum_enclosing_rectangle(vector<PT> &p) {
    int n = p.size();
    if (n <= 2) return perimeter(p);
    int mndot = 0; double tmp = dot(p[1] - p[0], p[0]);
    for (int i = 1; i < n; i++) {
        if (dot(p[1] - p[0], p[i]) <= tmp) {
            tmp = dot(p[1] - p[0], p[i]);
            mndot = i;
        }
    }
    double ans = inf;
    int i = 0, j = 1, mxdot = 1;
    while (i < n) {
        PT cur = p[(i + 1) % n] - p[i];
        while (cross(cur, p[(j + 1) % n] - p[j]) >= 0) j =
             (j + 1) % n;
        while (dot(p[(mxdot + 1) % n], cur) >=
             dot(p[mxdot], cur)) mxdot = (mxdot + 1) % n;
        while (dot(p[(mndot + 1) % n], cur) <=
             dot(p[mndot], cur)) mndot = (mndot + 1) % n;
        ans = min(ans, 2.0 * ((dot(p[mxdot], cur) /
             cur.norm() - dot(p[mndot], cur) / cur.norm())
             + dist_from_point_to_line(p[i], p[(i + 1) %
             n], p[j])));
        i++;
    }
    return ans;
}
```

3.15 MinkowskiSum

```
// a and b are strictly convex polygons of DISTINCT points
// returns a convex hull of their minkowski sum with
     distinct points
vector<PT> minkowski_sum(vector<PT> &a, vector<PT> &b) {
    int n = (int)a.size(), m = (int)b.size();
    int i = 0, j = 0; //assuming a[i] and b[j] both are
         (left, bottom)-most points
    vector<PT> c;
    c.push_back(a[i] + b[j]);
    while (1) {
        PT p1 = a[i] + b[(j + 1) % m];
        PT p2 = a[(i + 1) % n] + b[j];
        int t = orientation(c.back(), p1, p2);
        if (t >= 0) j = (j + 1) % m;
        if (t <= 0) i = (i + 1) % n, p1 = p2;
        if (t == 0) p1 = a[i] + b[j];
        if (p1 == c[0]) break;
        c.push_back(p1);
    }
    return c;
}
```

3.16 MonotoneChain

```
vector<Point> convex_hull(vector<Point> p, int n){
    sort(p.begin(), p.end(), [](const Point &A, const Point
         &B){
        return A.x != B.x ? A.x < B.x : A.y < B.y;
    });
    Point st = p[0], en = p[n - 1];
    vector<Point> up = {p[0]};
    vector<Point> down = {p[0]};
    for(int i = 1; i < n; ++i){
        if(i == n - 1 or cross(st, p[i], en) < 0){
            while((int)up.size() >= 2 and
                 cross(up[up.size() - 2], up.back(), p[i])
                 >= 0)
                up.pop_back();
            up.push_back(p[i]);
        }
        if(i == n - 1 or cross(st, p[i], en) > 0){
            while((int)down.size() >= 2 and
                 cross(down[down.size() - 2], down.back(),
                 p[i]) <= 0)
                down.pop_back();
            down.push_back(p[i]);
        }
    }
    p.clear();
    for(int i = 0; i < (int)up.size(); ++i)
        p.push_back(up[i]);
    for(int i = down.size() - 2; i >= 1; --i)
        p.push_back(down[i]);
    // return hull in clockwise order
    return p;
}
```

3.17 Point2D

```
const double inf = 1e100;
const double eps = 1e-9;
const double PI = acos((double)-1.0);
int sign(double x) { return (x > eps) - (x < -eps); }
struct PT {
```

```
    double x, y;
    PT() { x = 0, y = 0; }
    PT(double x, double y) : x(x), y(y) {}
    PT(const PT &p) : x(p.x), y(p.y) {}
    PT operator + (const PT &a) const { return PT(x + a.x,
         y + a.y); }
    PT operator - (const PT &a) const { return PT(x - a.x,
         y - a.y); }
    PT operator * (const double a) const { return PT(x * a,
         y * a); }
    friend PT operator * (const double &a, const PT &b) {
         return PT(a * b.x, a * b.y); }
    PT operator / (const double a) const { return PT(x / a,
         y / a); }
    bool operator == (PT a) const { return sign(a.x - x) ==
         0 && sign(a.y - y) == 0; }
    bool operator != (PT a) const { return !(*this == a); }
    bool operator < (PT a) const { return sign(a.x - x) ==
         0 ? y < a.y : x < a.x; }
    bool operator > (PT a) const { return sign(a.x - x) ==
         0 ? y > a.y : x > a.x; }
    double norm() { return sqrt(x * x + y * y); }
    double norm2() { return x * x + y * y; }
    PT perp() { return PT(-y, x); }
    double arg() { return atan2(y, x); }
    PT truncate(double r) { // returns a vector with norm r
         and having same direction
        double k = norm();
        if (!sign(k)) return *this;
        r /= k;
        return PT(x * r, y * r);
    }
};
inline double dot(PT a, PT b) { return a.x * b.x + a.y *
     b.y; }
inline double dist2(PT a, PT b) { return dot(a - b, a -
     b); }
inline double dist(PT a, PT b) { return sqrt(dot(a - b, a
     - b)); }
inline double cross(PT a, PT b) { return a.x * b.y - a.y *
     b.x; }
inline double cross2(PT a, PT b, PT c) { return cross(b -
     a, c - a); }
inline int orientation(PT a, PT b, PT c) { return
     sign(cross(b - a, c - a)); }
PT perp(PT a) { return PT(-a.y, a.x); }
PT rotateccw90(PT a) { return PT(-a.y, a.x); }
PT rotatecw90(PT a) { return PT(a.y, -a.x); }
PT rotateccw(PT a, double t) { return PT(a.x * cos(t) -
     a.y * sin(t), a.x * sin(t) + a.y * cos(t)); }
PT rotatecw(PT a, double t) { return PT(a.x * cos(t) + a.y
     * sin(t), -a.x * sin(t) + a.y * cos(t)); }
double SQ(double x) { return x * x; }
double rad_to_deg(double r) { return (r * 180.0 / PI); }
double deg_to_rad(double d) { return (d * PI / 180.0); }
double get_angle(PT a, PT b) {
    double costheta = dot(a, b) / a.norm() / b.norm();
    return acos(max((double)-1.0, min((double)1.0,
         costheta)));
}
bool is_point_in_angle(PT b, PT a, PT c, PT p) { // does
     point p lie in angle <bac
    assert(orientation(a, b, c) != 0);
    if (orientation(a, c, b) < 0) swap(b, c);
    return orientation(a, c, p) >= 0 && orientation(a, b,
         p) <= 0;
}
```

3.18 PointInsideHull

```
bool on_segment(const Point &A, const Point &B, const
     Point &C) { return cross(A, B, C) == 0 and dot(C, A,
     B) <= 0; }

bool check(vector<Point> &hull, Point &a) {
  int n = sz(hull);
  if (n == 1) return hull[0] == a;
  if (n == 2) return on_segment(hull[0], hull[1], a);
  if (cross(hull[0], hull[1], a) > 0) return 0;
  if (cross(hull[n - 1], hull[0], a) >= 0) return
       on_segment(hull[n - 1], hull[0], a);
  int l = 2, r = n - 1, ans = -1;
  while (l <= r) {
    int mid = (l + r) / 2;
    if (cross(hull[0], hull[mid], a) >= 0) {
      ans = mid;
      r = mid - 1;
    } else
      l = mid + 1;
  }
  debug(hull[0], hull[ans - 1], hull[ans], a, ans);
  return cross(hull[ans - 1], hull[ans], a) < 0 or
       on_segment(hull[ans - 1], hull[ans], a);
}
```

3.19 PolarSort

```
ll cross(P a, P b) { return a.first * b.second - a.second
     * b.first; }
int half(P p) {
       return (p.second > 0 || (p.second == 0 && p.first >
            0)) ? 0 : 1;
}
sort(v.begin(), v.end(), [](const P& a, const P& b) {
       int ha = half(a), hb = half(b);
       if (ha != hb) return ha < hb;
       return cross(a, b) > 0;
});
```

3.20 PolygonCircleIntersection

```
// intersection between a simple polygon and a circle
double polygon_circle_intersection(vector<PT> &v, PT p,
     double r) {
    int n = v.size();
    double ans = 0.00;
    PT org = {0, 0};
    for(int i = 0; i < n; i++) {
        int x = orientation(p, v[i], v[(i + 1) % n]);
        if(x == 0) continue;
        double area = triangle_circle_intersection(org, r,
             v[i] - p, v[(i + 1) % n] - p);
        if (x < 0) ans -= area;
        else ans += area;
    }
    return abs(ans);
}
```

3.21 PolygonDiameter

```
// Maximum distance of 2 points
double diameter(vector<PT> &p) {
    int n = (int)p.size();
    if (n == 1) return 0;
    if (n == 2) return dist(p[0], p[1]);
    double ans = 0;
    int i = 0, j = 1;
    while (i < n) {
        while (cross(p[(i + 1) % n] - p[i], p[(j + 1) % n]
             - p[j]) >= 0) {
            ans = max(ans, dist2(p[i], p[j]));
            j = (j + 1) % n;
        }
        ans = max(ans, dist2(p[i], p[j]));
        i++;
    }
    return sqrt(ans);
}
```

3.22 PolygonDistances

```
// minimum distance from a point to a convex polygon
// it assumes point lie strictly outside the polygon
double dist_from_point_to_polygon(vector<PT> &p, PT z) {
    double ans = inf;
    int n = p.size();
    if (n <= 3) {
        for(int i = 0; i < n; i++) ans = min(ans,
             dist_from_point_to_seg(p[i], p[(i + 1) % n],
             z));
        return ans;
    }
    auto [r, l] = tangents_from_point_to_polygon(p, z);
    if(l > r) r += n;
    while (l < r) {
        int mid = (l + r) >> 1;
        double left = dist2(p[mid % n], z), right=
             dist2(p[(mid + 1) % n], z);
        ans = min({ans, left, right});
        if(left < right) r = mid;
        else l = mid + 1;
```

```
    }
    ans = sqrt(ans);
    ans = min(ans, dist_from_point_to_seg(p[l % n], p[(l +
         1) % n], z));
    ans = min(ans, dist_from_point_to_seg(p[l % n], p[(l -
         1 + n) % n], z));
    return ans;
}
// minimum distance from convex polygon p to line ab
// returns 0 is it intersects with the polygon
// top - upper right vertex
double dist_from_polygon_to_line(vector<PT> &p, PT a, PT
     b, int top) { //O(log n)
    PT orth = (b - a).perp();
    if (orientation(a, b, p[0]) > 0) orth = (a - b).perp();
    int id = extreme_vertex(p, orth, top);
    if (dot(p[id] - a, orth) > 0) return 0.0; //if orth and
         a are in the same half of the line, then poly and
         line intersects
    return dist_from_point_to_line(a, b, p[id]); //does not
         intersect
}
// minimum distance from a convex polygon to another
     convex polygon
// the polygon doesnot overlap or touch
// tested in https://toph.co/p/the-wall
double dist_from_polygon_to_polygon(vector<PT> &p1,
     vector<PT> &p2) { // O(n log n)
    double ans = inf;
    for (int i = 0; i < p1.size(); i++) {
        ans = min(ans, dist_from_point_to_polygon(p2,
             p1[i]));
    }
    for (int i = 0; i < p2.size(); i++) {
        ans = min(ans, dist_from_point_to_polygon(p1,
             p2[i]));
    }
    return ans;
}
// maximum distance from a convex polygon to another
     convex polygon
double maximum_dist_from_polygon_to_polygon(vector<PT> &u,
     vector<PT> &v){ //O(n)
    int n = (int)u.size(), m = (int)v.size();
    double ans = 0;
    if (n < 3 || m < 3) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < m; j++) ans = max(ans,
                 dist2(u[i], v[j]));
        }
        return sqrt(ans);
    }
    if (u[0].x > v[0].x) swap(n, m), swap(u, v);
    int i = 0, j = 0, step = n + m + 10;
    while (j + 1 < m && v[j].x < v[j + 1].x) j++ ;
    while (step--) {
        if (cross(u[(i + 1)%n] - u[i], v[(j + 1)%m] - v[j])
             >= 0) j = (j + 1) % m;
        else i = (i + 1) % n;
        ans = max(ans, dist2(u[i], v[j]));
    }
    return sqrt(ans);
}
```

3.23 PolygonLineIntersection

```
// not necessarily convex, boundary is included in the
     intersection
// returns total intersected length
double polygon_line_intersection(vector<PT> p, PT a, PT b)
     {
    int n = p.size();
    p.push_back(p[0]);
    line l = line(a, b);
    double ans = 0.0;
    vector< pair<double, int> > vec;
    for (int i = 0; i < n; i++) {
        int s1 = sign(cross(b - a, p[i] - a));
        int s2 = sign(cross(b - a, p[i+1] - a));
        if (s1 == s2) continue;
        line t = line(p[i], p[i + 1]);
        PT inter = (t.v * l.c - l.v * t.c) / cross(l.v,
             t.v);
        double tmp = dot(inter, l.v);
        int f;
        if (s1 > s2) f = s1 && s2 ? 2 : 1;
        else f = s1 && s2 ? -2 : -1;
        vec.push_back(make_pair(tmp, f));
    }
    sort(vec.begin(), vec.end());
    for (int i = 0, j = 0; i + 1 < (int)vec.size(); i++){
        j += vec[i].second;
        if (j) ans += vec[i + 1].first - vec[i].first;
    }
    ans = ans / sqrt(dot(l.v, l.v));
    p.pop_back();
    return ans;
}
```

3.24 PolygonUnion

```
// calculates the area of the union of n polygons (not
     necessarily convex).
// the points within each polygon must be given in CCW
     order.
// complexity: O(N^2), where N is the total number of
     points
double rat(PT a, PT b, PT p) {
       return !sign(a.x - b.x) ? (p.y - a.y) / (b.y - a.y)
            : (p.x - a.x) / (b.x - a.x);
 };
double polygon_union(vector<vector<PT>> &p) {
    int n = p.size();
    double ans=0;
    for(int i = 0; i < n; ++i) {
        for (int v = 0; v < (int)p[i].size(); ++v) {
            PT a = p[i][v], b = p[i][(v + 1) % p[i].size()];
            vector<pair<double, int>> segs;
            segs.emplace_back(0, 0), segs.emplace_back(1,
                 0);
            for(int j = 0; j < n; ++j) {
                if(i != j) {
                    for(size_t u = 0; u < p[j].size(); ++u) {
                        PT c = p[j][u], d = p[j][(u + 1) %
                             p[j].size()];
                        int sc = sign(cross(b - a, c - a)),
                             sd = sign(cross(b - a, d - a));
                        if(!sc && !sd) {
                            if(sign(dot(b - a, d - c)) > 0 &&
                                 i > j) {
                                segs.emplace_back(rat(a, b,
                                     c), 1),
                                     segs.emplace_back(rat(a,
                                     b, d), -1);
                            }
                        }
                        else {
                            double sa = cross(d - c, a - c),
                                 sb = cross(d - c, b - c);
                            if(sc >= 0 && sd < 0)
                                segs.emplace_back(sa / (sa -
                                     sb), 1);
                            else if(sc < 0 && sd >= 0)
                                segs.emplace_back(sa / (sa -
                                     sb), -1);
                        }
                    }
                }
            }
            sort(segs.begin(), segs.end());
            double pre = min(max(segs[0].first, 0.0), 1.0),
                 now, sum = 0;
            int cnt = segs[0].second;
            for(int j = 1; j < segs.size(); ++j) {
                now = min(max(segs[j].first, 0.0), 1.0);
                if (!cnt) sum += now - pre;
                cnt += segs[j].second;
                pre = now;
            }
            ans += cross(a, b) * sum;
        }
    }
    return ans * 0.5;
}
```

3.25 PolygonWidth

```
// Maximum distance between 2 points IN the polygon
double width(vector<PT> &p) {
    int n = (int)p.size();
    if (n <= 2) return 0;
    double ans = inf;
    int i = 0, j = 1;
    while (i < n) {
        while (cross(p[(i + 1) % n] - p[i], p[(j + 1) % n]
             - p[j]) >= 0) j = (j + 1) % n;
```

```
        ans = min(ans, dist_from_point_to_line(p[i], p[(i +
            1) % n], p[j]));
        i++;
    }
    return ans;
}
```

3.26 Segment

```
// returns true if point p is on line segment ab
bool is_point_on_seg(PT a, PT b, PT p) {
    if (fabs(cross(p - b, a - b)) < eps) {
        if (p.x < min(a.x, b.x) || p.x > max(a.x, b.x))
            return false;
        if (p.y < min(a.y, b.y) || p.y > max(a.y, b.y))
            return false;
        return true;
    }
    return false;
}
// minimum distance point from point c to segment ab that
    lies on segment ab
PT project_from_point_to_seg(PT a, PT b, PT c) {
    double r = dist2(a, b);
    if (sign(r) == 0) return a;
    r = dot(c - a, b - a) / r;
    if (r < 0) return a;
    if (r > 1) return b;
    return a + (b - a) * r;
}
// minimum distance from point c to segment ab
double dist_from_point_to_seg(PT a, PT b, PT c) {
    return dist(c, project_from_point_to_seg(a, b, c));
}
// intersection point between segment ab and segment cd
    assuming unique intersection exists
bool seg_seg_intersection(PT a, PT b, PT c, PT d, PT &ans)
    {
    double oa = cross2(c, d, a), ob = cross2(c, d, b);
    double oc = cross2(a, b, c), od = cross2(a, b, d);
    if (oa * ob < 0 && oc * od < 0){
        ans = (a * ob - b * oa) / (ob - oa);
        return 1;
    }
    else return 0;
}
// intersection point between segment ab and segment cd
    assuming unique intersection may not exists
// se.size()==0 means no intersection
// se.size()==1 means one intersection
// se.size()==2 means range intersection
set<PT> seg_seg_intersection_inside(PT a, PT b, PT c, PT
    d) {
    PT ans;
    if (seg_seg_intersection(a, b, c, d, ans)) return {ans};
    set<PT> se;
    if (is_point_on_seg(c, d, a)) se.insert(a);
    if (is_point_on_seg(c, d, b)) se.insert(b);
    if (is_point_on_seg(a, b, c)) se.insert(c);
    if (is_point_on_seg(a, b, d)) se.insert(d);
    return se;
}
// intersection between segment ab and line cd
// 0 if do not intersect, 1 if proper intersect, 2 if
    segment intersect
int seg_line_relation(PT a, PT b, PT c, PT d) {
    double p = cross2(c, d, a);
    double q = cross2(c, d, b);
    if (sign(p) == 0 && sign(q) == 0) return 2;
    else if (p * q < 0) return 1;
    else return 0;
}
// intersection between segament ab and line cd assuming
    unique intersection exists
bool seg_line_intersection(PT a, PT b, PT c, PT d, PT
    &ans) {
    bool k = seg_line_relation(a, b, c, d);
    assert(k != 2);
    if (k) line_line_intersection(a, b, c, d, ans);
    return k;
}
// minimum distance from segment ab to segment cd
double dist_from_seg_to_seg(PT a, PT b, PT c, PT d) {
    PT dummy;
    if (seg_seg_intersection(a, b, c, d, dummy)) return 0.0;
    else return min({dist_from_point_to_seg(a, b, c),
        dist_from_point_to_seg(a, b, d),
        dist_from_point_to_seg(c, d, a),
        dist_from_point_to_seg(c, d, b)});
}
```

3.27 SmallestEnclosingCircle

```
double eps = 1e-9;
using Point = complex<double>;
struct Circle{ Point p; double r; };
double dist(Point p, Point q){ return abs(p-q); }
double area2(Point p, Point q){ return (conj(p)*q).imag();}
bool in(const Circle& c, Point p){ return dist(c.p, p) <
    c.r + eps; }
Circle INVAL = Circle{Point(0, 0), -1};
Circle mCC(Point a, Point b, Point c){
    b -= a; c -= a;
    double d = 2*(conj(b)*c).imag(); if(abs(d)<eps)
        return INVAL;
    Point ans = (c*norm(b) - b*norm(c)) * Point(0, -1)
        / d;
    return Circle{a + ans, abs(ans)};
}
Circle solve(vector<Point> p) {
    mt19937 gen(0x94949); shuffle(p.begin(), p.end(),
        gen);
    Circle c = INVAL;
    for(int i=0; i<p.size(); ++i) if(c.r<0 ||!in(c,
        p[i])){
        c = Circle{p[i], 0};
        for(int j=0; j<=i; ++j) if(!in(c, p[j])){
            Circle ans{(p[i]+p[j])*0.5,
                dist(p[i], p[j])*0.5};
            if(c.r == 0) {c = ans; continue;}
            Circle l, r; l = r = INVAL;
            Point pq = p[j]-p[i];
            for(int k=0; k<=j; ++k) if(!in(ans,
                p[k])) {
                double a2 = area2(pq,
                    p[k]-p[i]);
                Circle c = mCC(p[i], p[j],
                    p[k]);
                if(c.r<0) continue;
                else if(a2 > 0 &&
                    (l.r<0||area2(pq,
                    c.p-p[i]) > area2(pq,
                    l.p-p[i]))) l = c;
                else if(a2 < 0 &&
                    (r.r<0||area2(pq,
                    c.p-p[i]) < area2(pq,
                    r.p-p[i]))) r = c;
            }
            if(l.r<0&&r.r<0) c = ans;
            else if(l.r<0) c = r;
            else if(r.r<0) c = l;
            else c = l.r<=r.r?l:r;
        }
    }
    return c;
}
```

3.28 TriangleCircleIntersection

```
// system should be translated from circle center
double triangle_circle_intersection(PT c, double r, PT a,
    PT b) {
    double sd1 = dist2(c, a), sd2 = dist2(c, b);
    if(sd1 > sd2) swap(a, b), swap(sd1, sd2);
    double sd = dist2(a, b);
    double d1 = sqrtl(sd1), d2 = sqrtl(sd2), d = sqrt(sd);
    double x = abs(sd2 - sd - sd1) / (2 * d);
    double h = sqrtl(sd1 - x * x);
    if(r >= d2) return h * d / 2;
    double area = 0;
    if(sd + sd1 < sd2) {
        if(r < d1) area = r * r * (acos(h / d2) - acos(h /
            d1)) / 2;
        else {
            area = r * r * ( acos(h / d2) - acos(h / r)) /
                2;
            double y = sqrtl(r * r - h * h);
            area += h * (y - x) / 2;
        }
    }
    else {
        if(r < h) area = r * r * (acos(h / d2) + acos(h /
            d1)) / 2;
        else {
            area += r * r * (acos(h / d2) - acos(h / r)) /
                2;
```

```
            double y = sqrtl(r * r - h * h);
            area += h * y / 2;
            if(r < d1) {
                area += r * r * (acos(h / d1) - acos(h / r))
                     / 2;
                area += h * y / 2;
            }
            else area += h * x / 2;
        }
    }
    return area;
}
```

3.29 Utilities

```
double perimeter(vector<PT> &p) {
    double ans=0; int n = p.size();
    for (int i = 0; i < n; i++) ans += dist(p[i], p[(i + 1)
         % n]);
    return ans;
}
double area(vector<PT> &p) {
    double ans = 0; int n = p.size();
    for (int i = 0; i < n; i++) ans += cross(p[i], p[(i +
         1) % n]);
    return fabs(ans) * 0.5;
}
double area_of_triangle(PT a, PT b, PT c) {
    return fabs(cross(b - a, c - a) * 0.5);
}
// 0 if cw, 1 if ccw
bool get_direction(vector<PT> &p) {
    double ans = 0; int n = p.size();
    for (int i = 0; i < n; i++) ans += cross(p[i], p[(i +
         1) % n]);
    if (sign(ans) > 0) return 1;
    return 0;
}
// find a point from a through b with distance d
PT point_along_line(PT a, PT b, double d) {
    assert(a != b);
    return a + (((b - a) / (b - a).norm()) * d);
}
// projection point c onto line through a and b assuming a
     != b
PT project_from_point_to_line(PT a, PT b, PT c) {
    return a + (b - a) * dot(c - a, b - a) / (b -
         a).norm2();
}
// reflection point c onto line through a and b assuming a
     != b
PT reflection_from_point_to_line(PT a, PT b, PT c) {
    PT p = project_from_point_to_line(a,b,c);
    return p + p - c;
}
// minimum distance from point c to line through a and b
double dist_from_point_to_line(PT a, PT b, PT c) {
    return fabs(cross(b - a, c - a) / (b - a).norm());
}
```

```
// 0 if not parallel, 1 if parallel, 2 if collinear
int is_parallel(PT a, PT b, PT c, PT d) {
    double k = fabs(cross(b - a, d - c));
    if (k < eps){
        if (fabs(cross(a - b, a - c)) < eps && fabs(cross(c
             - d, c - a)) < eps) return 2;
        else return 1;
    }
    else return 0;
}
// check if two lines are same
bool are_lines_same(PT a, PT b, PT c, PT d) {
    if (fabs(cross(a - c, c - d)) < eps && fabs(cross(b -
         c, c - d)) < eps) return true;
    return false;
}

// 1 if point is ccw to the line, 2 if point is cw to the
     line, 3 if point is on the line
int point_line_relation(PT a, PT b, PT p) {
    int c = sign(cross(p - a, b - a));
    if (c < 0) return 1;
    if (c > 0) return 2;
    return 3;
}
```

4 Graph

4.1 2SAT

```
// solve (a or not b) and (not c or not d) and (not a or
     c) and ...
struct TwoSatSolver {
  int n_vars;
  int n_vertices;
  vector<vector<int>> adj, adj_t;
  vector<bool> used;
  vector<int> order, comp;
  vector<bool> assignment;
  TwoSatSolver(int _n_vars)
      : n_vars(_n_vars),
        n_vertices(2 * n_vars),
        adj(n_vertices),
        adj_t(n_vertices),
        used(n_vertices),
        order(),
        comp(n_vertices, -1),
        assignment(n_vars) {
    order.reserve(n_vertices);
  }
  void dfs1(int v) {
    used[v] = true;
    for (int u : adj[v]) {
      if (!used[u]) dfs1(u);
    }
    order.push_back(v);
  }
  void dfs2(int v, int cl) {
    comp[v] = cl;
    for (int u : adj_t[v]) {
      if (comp[u] == -1) dfs2(u, cl);
    }
  }
  bool solve_2SAT() {
    order.clear();
    used.assign(n_vertices, false);
    for (int i = 0; i < n_vertices; ++i) {
      if (!used[i]) dfs1(i);
    }

    comp.assign(n_vertices, -1);
    for (int i = 0, j = 0; i < n_vertices; ++i) {
      int v = order[n_vertices - i - 1];
      if (comp[v] == -1) dfs2(v, j++);
    }

    assignment.assign(n_vars, false);
    for (int i = 0; i < n_vertices; i += 2) {
      if (comp[i] == comp[i + 1]) return false;
      assignment[i / 2] = comp[i] > comp[i + 1];
    }
    return true;
  }
  void add_disjunction(int a, bool na, int b, bool nb) {
    // na and nb signify whether a and b are to be negated
    a = 2 * a ^ na;
    b = 2 * b ^ nb;
    int neg_a = a ^ 1;
    int neg_b = b ^ 1;
    adj[neg_a].push_back(b);
    adj[neg_b].push_back(a);
    adj_t[b].push_back(neg_a);
    adj_t[a].push_back(neg_b);
  }
  static void example_usage() {
    TwoSatSolver solver(3);                    // a, b, c
    solver.add_disjunction(0, false, 1, true); //  a  v
         not b
    solver.add_disjunction(0, true, 1, true); // not a v
         not b
    solver.add_disjunction(1, false, 2, false); // b  v
         c
    solver.add_disjunction(0, false, 0, false); // a  v
         a
    assert(solver.solve_2SAT() == true);
    auto expected = vector<bool>(True, False, True);
    assert(solver.assignment == expected);
  }
};
```

4.2 BlockCutTree

```
int n, m, q;
vector<int> g[mxN];
int low[mxN], num[mxN], timer;
vector<vector<int>> comps;
bool is_cutpoint[mxN];
```

```
vector<int> stk;
vector<int> blockcut_tree[mxN];
int up[mxN][17], depth[mxN], id[mxN];

namespace biconnected_components {
void dfs(int u, int fa) {
	int nchild = 0;
	if (!g[u].size()) {
		is_cutpoint[u] = 1;
		comps.push_back({u});
		return;
	}
	num[u] = low[u] = ++timer;
	stk.push_back(u);
	for (int v : g[u]) {
		if (v == fa) continue;
		if (num[v])
			low[u] = min(low[u], num[v]);
		else {
			dfs(v, u);
			low[u] = min(low[u], low[v]);
			nchild++;
			if (low[v] >= num[u]) {
				is_cutpoint[u] = (u != fa ||
					nchild > 1);
				comps.push_back({u});
				while (comps.back().back() !=
					v) {
					comps.back().push_back(stk.back());
					stk.pop_back();
				}
			}
		}
	}
}

void build_tree() {
	int node_id = 0;
	for (int i = 1; i <= n; i++) {
		if (is_cutpoint[i]) {
			id[i] = ++node_id;
		}
	}
	for (auto& comp : comps) {
		int node = ++node_id;
		for (int u : comp) {
			if (!is_cutpoint[u])
				id[u] = node;
			else {
				blockcut_tree[node].push_back(id[u]);
				blockcut_tree[id[u]].push_back(node);
			}
		}
	}
}
} // namespace biconnected_components
```

4.3 CartesianTree

```
// build min Cartesian tree
std::vector<int> lc(n, n), rc(n, n);
std::vector<int> stk;
for (int i = 0; i < n; i++) {
    while (!stk.empty() && b[i] < b[stk.back()]) { // flip
        to '>' to get max Cartesian tree
        rc[stk.back()] = lc[i];
        lc[i] = stk.back();
        stk.pop_back();
    }
    stk.push_back(i);
}

while (stk.size() > 1) {
    int x = stk.back();
    stk.pop_back();
    rc[stk.back()] = x;
}
```

4.4 Dinic

```
const ll INF = 1e18;
struct Dinic {
  const static bool SCALING = false; // scaling = EV
      log(max C) with larger constant
ll lim = 1;
  struct Edge {
    int u, v;
    ll cap, flow;
  };
  int n, s, t;
  vector<int> level, ptr;
  vector<Edge> e;
  vector<vector<int>> g;
  Dinic(int _n) : n(_n), level(_n), ptr(_n), g(_n) {
    e.clear();
    for (int i = 0; i < n; ++i) {
      ptr[i] = 0;
      g[i].clear();
    }
  }
  void add_edge(int u, int v, ll c) {
    debug(u, v, c);
    g[u].push_back(sz(e));
    e.push_back({u, v, c, 0});
    g[v].push_back(sz(e));
    e.push_back({v, u, 0, 0});
  }
  ll get_max_flow(int _s, int _t) {
    s = _s, t = _t;
    ll flow = 0;
    for (lim = SCALING ? (1 << 30) : 1; lim > 0; lim >>= 1)
        {
      while (1) {
        if (!bfs()) break;
        fill(all(ptr), 0);
        while (ll pushed = dfs(s, INF)) flow += pushed;
      }
    }
    return flow;
  }
 private:
  bool bfs() {
    queue<int> q;
    q.push(s);
    fill(all(level), -1);
    level[s] = 0;
    while (!q.empty()) {
      int u = q.front();
      q.pop();
      for (int id : g[u]) {
        if (e[id].cap - e[id].flow < 1) continue;
        if (level[e[id].v] != -1) continue;
        if (SCALING and e[id].cap - e[id].flow < lim)
            continue;
        level[e[id].v] = level[u] + 1;
        q.push(e[id].v);
      }
    }
    return level[t] != -1;
  }
  ll dfs(int u, ll flow) {
    if (!flow) return 0;
    if (u == t) return flow;
    for (; ptr[u] < sz(g[u]); ++ptr[u]) {
      int id = g[u][ptr[u]], to = e[id].v;
      if (level[to] != level[u] + 1) continue;
      ll pushed = dfs(to, min(flow, e[id].cap -
          e[id].flow));
      if (pushed) {
        e[id].flow += pushed;
        e[id ^ 1].flow -= pushed;
        return pushed;
      }
    }
    return 0;
  }
};
```

4.5 EulerPath

```
struct EulerUndirected {
  EulerUndirected(int _n) : n(_n), m(0), adj(_n), deg(_n,
      0) {}
  void add_edge(int u, int v) {
    adj[u].push_front(Edge(v));
    auto it1 = adj[u].begin();
    adj[v].push_front(Edge(u));
    auto it2 = adj[v].begin();

    it1->rev = it2;
    it2->rev = it1;

    ++deg[u];
    ++deg[v];
    ++m;
  }
  pair<bool, vector<int>> solve() {
```

```
    int cntOdd = 0;
    int start = -1;
    for (int i = 0; i < n; i++) {
      if (deg[i] % 2) {
        ++cntOdd;
        if (cntOdd > 2) return {false, {}};

        if (start < 0) start = i;
      }
    }
    // no odd vertex -> start from any vertex with positive
        degree
    if (start < 0) {
      for (int i = 0; i < n; i++) {
        if (deg[i]) {
          start = i;
          break;
        }
      }
      if (start < 0) {
        // no edge -> empty path
        return {true, {}};
      }
    }

    vector<int> path;
    find_path(start, path);

    if (m + 1 != static_cast<int>(path.size())) {
      return {false, {}};
    }

    return {true, path};
  }
  struct Edge {
    int to;
    list<Edge>::iterator rev;

    Edge(int _to) : to(_to) {}
  };
  // private:
  int n, m;
  vector<list<Edge>> adj;
  vector<int> deg;

  void find_path(int v, vector<int>& path) {
    while (adj[v].size() > 0) {
      int next = adj[v].front().to;
      adj[next].erase(adj[v].front().rev);
      adj[v].pop_front();
      find_path(next, path);
    }
    path.push_back(v);
  }
};
```

4.6 EulerPathDirected

```
struct EulerDirected {
 EulerDirected(int _n) : n(_n), adj(n), in_deg(n, 0),
      out_deg(n, 0) {}
 void add_edge(int u, int v) { // directed edge
   assert(0 <= u && u < n);
   assert(0 <= v && v < n);
   adj[u].push_front(v);
   in_deg[v]++;
   out_deg[u]++;
 }
 pair<bool, vector<int>> solve() {
   int start = -1, last = -1;
   for (int i = 0; i < n; i++) {
     // for all u, |in_deg(u) - out_deg(u)| <= 1
     if (abs(in_deg[i] - out_deg[i]) > 1) return {false,
          {}};

     if (out_deg[i] > in_deg[i]) {
       // At most 1 vertex with out_deg[u] - in_deg[u] = 1
            (start vertex)
       if (start >= 0) return {false, {}};
       start = i;
     }

     if (in_deg[i] > out_deg[i]) {
       // At most 1 vertex with in_deg[u] - out_deg[u] = 1
            (last vertex)
       if (last >= 0) return {false, {}};
       last = i;
     }
   }

   // can start at any vertex with degree > 0
   if (start < 0) {
     for (int i = 0; i < n; i++) {
       if (in_deg[i]) {
         start = i;
         break;
       }
     }
     // no start vertex --> all vertices have degree == 0
     if (start < 0) return {true, {}};
   }
   vector<int> path;
   find_path(start, path);
   reverse(path.begin(), path.end());

   // check that we visited all vertices with degree > 0
   vector<bool> visited(n, false);
   for (int u : path) visited[u] = true;

   for (int u = 0; u < n; u++) {
     if (in_deg[u] && !visited[u]) {
       return {false, {}};
     }
   }

   return {true, path};
 }
private:
 int n;
 vector<list<int>> adj;
 vector<int> in_deg, out_deg;
 void find_path(int v, vector<int>& path) {
   while (adj[v].size() > 0) {
     int next = adj[v].front();
     adj[v].pop_front();
     find_path(next, path);
   }
   path.push_back(v);
 }
};
```

4.7 GeneralMatching

```
const int MAXN = 2020 + 1;
struct GM { // 1-based Vertex index
  int vis[MAXN], par[MAXN], orig[MAXN], match[MAXN],
      aux[MAXN], t, N;
  vector<int> conn[MAXN];
  queue<int> Q;
  void addEdge(int u, int v) {
    conn[u].push_back(v);
    conn[v].push_back(u);
  }
  void init(int n) {
    N = n;
    t = 0;
    for (int i = 0; i <= n; ++i) {
      conn[i].clear();
      match[i] = aux[i] = par[i] = 0;
    }
  }
  void augment(int u, int v) {
    int pv = v, nv;
    do {
      pv = par[v];
      nv = match[pv];
      match[v] = pv;
      match[pv] = v;
      v = nv;
    } while (u != pv);
  }
  int lca(int v, int w) {
    ++t;
    while (true) {
      if (v) {
        if (aux[v] == t) return v;
        aux[v] = t;
        v = orig[par[match[v]]];
      }
      swap(v, w);
    }
  }
  void blossom(int v, int w, int a) {
    while (orig[v] != a) {
      par[v] = w;
      w = match[v];
      if (vis[w] == 1) Q.push(w), vis[w] = 0;
      orig[v] = orig[w] = a;
      v = par[w];
```

```
  }
 }
 bool bfs(int u) {
  fill(vis + 1, vis + 1 + N, -1);
  iota(orig + 1, orig + N + 1, 1);
  Q = queue<int>();
  Q.push(u);
  vis[u] = 0;
  while (!Q.empty()) {
    int v = Q.front();
    Q.pop();
    for (int x : conn[v]) {
      if (vis[x] == -1) {
        par[x] = v;
        vis[x] = 1;
        if (!match[x]) return augment(u, x), true;
        Q.push(match[x]);
        vis[match[x]] = 0;
      } else if (vis[x] == 0 && orig[v] != orig[x]) {
        int a = lca(orig[v], orig[x]);
        blossom(x, v, a);
        blossom(v, x, a);
      }
    }
  }
  return false;
 }
 int Match() {
  int ans = 0;
  // find random matching (not necessary, constant
       improvement)
  vector<int> V(N - 1);
  iota(V.begin(), V.end(), 1);
  shuffle(V.begin(), V.end(), mt19937(0x94949));
  for (auto x : V)
    if (!match[x]) {
      for (auto y : conn[x])
        if (!match[y]) {
          match[x] = y, match[y] = x;
          ++ans;
          break;
        }
    }
  for (int i = 1; i <= N; ++i)
    if (!match[i] && bfs(i)) ++ans;
  return ans;
 }
};
```

4.8 GlobalMinCut

```
pair<int, vi> GetMinCut(vector<vi>& weights) {
  int N = sz(weights);
  vi used(N), cut, best_cut;
  int best_weight = -1;
  for (int phase = N - 1; phase >= 0; phase--) {
    vi w = weights[0], added = used;
    int prev, k = 0;
    rep(i, 0, phase) {
      prev = k;
      k = -1;
      rep(j, 1, N) if (!added[j] && (k == -1 || w[j] >
           w[k])) k = j;
      if (i == phase - 1) {
        rep(j, 0, N) weights[prev][j] += weights[k][j];
        rep(j, 0, N) weights[j][prev] = weights[prev][j];
        used[k] = true;
        cut.push_back(k);
        if (best_weight == -1 || w[k] < best_weight) {
          best_cut = cut;
          best_weight = w[k];
        }
      } else {
        rep(j, 0, N) w[j] += weights[k][j];
        added[k] = true;
      }
    }
  }
  return {best_weight, best_cut};
}
```

4.9 HopcroftKarp

```
struct maximum_bipartite_matching {
    int n, m, turn, matched;
    vector<int> matchR, matchL, vis;
    vector< vector<int> > adj;
    vector< pair<int, int> > edges;
    maximum_bipartite_matching(int _n, int _m): n(_n),
         m(_m), matchL(n + 1), matchR(m + 1, -1), vis(n +
         1), adj(n + 1), turn(0), matched(0) {
    }

    void add_edge(int u, int v) {
        edges.push_back({u, v});
    }

    bool dfs(int u) {
        if (vis[u] == turn) return false;
        vis[u] = turn;

        for(int v: adj[u]) {
            if (matchR[v] == -1 || dfs(matchR[v])) {
                matchR[v] = u;
                return true;
            }
        }
        return false;
    }

    void match() {
        shuffle(edges.begin(), edges.end(), rng);
        for(pair<int, int> edge: edges) {
            adj[edge.first].push_back(edge.second);
        }
        bool flag = 1;

        while(flag) {
            flag = 0;
            ++turn;
            for(int u = 0; u < n; ++u) {
                if (matchL[u])
                    continue;
                if (dfs(u)) {
                    flag = matchL[u] = true;
                    ++matched;
                }
            }
        }
    }

    vector< pair<int, int> > get_edges() {
        vector< pair<int, int> > res;
        for(int v = 0; v < m; ++v) {
            if (matchR[v] != -1) {
                res.push_back({matchR[v], v});
            }
        }
        return res;
    }
};
```

4.10 JointsnBridges

```
const int maxN = 10010;
int n, m;
bool joint[maxN];
int timeDfs = 0, bridge = 0;
int low[maxN], num[maxN];
vector <int> g[maxN];
 void dfs(int u, int pre) {
  int child = 0;
  num[u] = low[u] = ++timeDfs;
  for (int v : g[u]) {
    if (v == pre) continue;
    if (!num[v]) {
      dfs(v, u);
      low[u] = min(low[u], low[v]);
      if (low[v] == num[v]) bridge++;
      child++;
      if (u == pre) { // if u is root of DFS tree
        if (child > 1) joint[u] = true;
      }
      else if (low[v] >= num[u]) joint[u] = true;
    }
    else low[u] = min(low[u], num[v]);
  }
}
int main() {
  cin >> n >> m;
  for (int i = 1; i <= m; i++) {
    int u, v;
    cin >> u >> v;
    g[u].push_back(v);
    g[v].push_back(u);
  }
  for (int i = 1; i <= n; i++)
```

```
    if (!num[i]) dfs(i, i);
  int cntJoint = 0;
  for (int i = 1; i <= n; i++) cntJoint += joint[i];
  cout << cntJoint << ' ' << bridge;
}
```

4.11 MCMF slow

```
// Time Complexity: O(F * (V * E))
template <typename T, typename C>
class mcmf {
 public:
  static constexpr T eps = (T) 1e-9;

  struct edge {
    int from;
    int to;
    T c;
    T f;
    C cost;
  };

  vector<vector<int>> g;
  vector<edge> edges;
  vector<C> d;
  vector<int> q;
  vector<bool> in_queue;
  vector<int> pe;
  int n;
  int st, fin;
  T flow;
  C cost;

  mcmf(int _n, int _st, int _fin) : n(_n), st(_st),
       fin(_fin) {
    assert(0 <= st && st < n && 0 <= fin && fin < n && st
         != fin);
    g.resize(n);
    d.resize(n);
    in_queue.resize(n);
    pe.resize(n);
    flow = 0;
    cost = 0;
  }

  void clear_flow() {
    for (const edge &e : edges) {
      e.f = 0;
    }
    flow = 0;
  }

  void add(int from, int to, T forward_cap, T
       backward_cap, C cost) {
    assert(0 <= from && from < n && 0 <= to && to < n);
    g[from].push_back((int) edges.size());
    edges.push_back({from, to, forward_cap, 0, cost});
    g[to].push_back((int) edges.size());
    edges.push_back({to, from, backward_cap, 0, -cost});
  }

  bool expath() {
    fill(d.begin(), d.end(), numeric_limits<C>::max());
    q.clear();
    q.push_back(st);
    d[st] = 0;
    in_queue[st] = true;
    int beg = 0;
    bool found = false;
    while (beg < (int) q.size()) {
      int i = q[beg++];
      if (i == fin) {
        found = true;
      }
      in_queue[i] = false;
      for (int id : g[i]) {
        const edge &e = edges[id];
        if (e.c - e.f > eps && d[i] + e.cost < d[e.to]) {
          d[e.to] = d[i] + e.cost;
          pe[e.to] = id;
          if (!in_queue[e.to]) {
            q.push_back(e.to);
            in_queue[e.to] = true;
          }
        }
      }
    }
    if (found) {
      T push = numeric_limits<T>::max();
      int v = fin;
      while (v != st) {
        const edge &e = edges[pe[v]];
        push = min(push, e.c - e.f);
        v = e.from;
      }
      v = fin;
      while (v != st) {
        edge &e = edges[pe[v]];
        e.f += push;
        edge &back = edges[pe[v] ^ 1];
        back.f -= push;
        v = e.from;
      }
      flow += push;
      cost += push * d[fin];
    }
    return found;
  }

  pair<T, C> max_flow_min_cost() {
    while (expath()) {}
    return {flow, cost};
  }
};
```

4.12 MCMF

```
// Time Complexity: O(F * (E * log V))
template <typename T, typename C>
class MCMF {
 public:
  static constexpr T eps = (T) 1e-9;

  struct edge {
    int from;
    int to;
    T c;
    T f;
    C cost;
  };

  int n;
  vector<vector<int>> g;
  vector<edge> edges;
  vector<C> d;
  vector<C> pot;
  __gnu_pbds::priority_queue<pair<C, int>> q;
  vector<typename decltype(q)::point_iterator> its;
  vector<int> pe;
  const C INF_C = numeric_limits<C>::max() / 2;

  explicit MCMF(int n_) : n(n_), g(n), d(n), pot(n, 0),
       its(n), pe(n) {}

  int add(int from, int to, T forward_cap, T backward_cap,
       C edge_cost) {
    assert(0 <= from && from < n && 0 <= to && to < n);
    assert(forward_cap >= 0 && backward_cap >= 0);
    int id = static_cast<int>(edges.size());
    g[from].push_back(id);
    edges.push_back({from, to, forward_cap, 0, edge_cost});
    g[to].push_back(id + 1);
    edges.push_back({to, from, backward_cap, 0,
         -edge_cost});
    return id;
  }

  void expath(int st) {
    fill(d.begin(), d.end(), INF_C);
    q.clear();
    fill(its.begin(), its.end(), q.end());
    its[st] = q.push({pot[st], st});
    d[st] = 0;
    while (!q.empty()) {
      int i = q.top().second;
      q.pop();
      its[i] = q.end();
      for (int id : g[i]) {
        const edge &e = edges[id];
        int j = e.to;
        if (e.c - e.f > eps && d[i] + e.cost < d[j]) {
          d[j] = d[i] + e.cost;
          pe[j] = id;
          if (its[j] == q.end()) {
            its[j] = q.push({pot[j] - d[j], j});
          } else {
            q.modify(its[j], {pot[j] - d[j], j});
          }
        }
      }
```

```
  }
  swap(d, pot);
}

pair<T, C> max_flow_min_cost(int st, int fin) {
  T flow = 0;
  C cost = 0;
  bool ok = true;
  for (auto& e : edges) {
    if (e.c - e.f > eps && e.cost + pot[e.from] -
        pot[e.to] < 0) {
      ok = false;
      break;
    }
  }
  if (ok) {
    expath(st);
  } else {
    vector<int> deg(n, 0);
    for (int i = 0; i < n; i++) {
      for (int eid : g[i]) {
        auto& e = edges[eid];
        if (e.c - e.f > eps) {
          deg[e.to] += 1;
        }
      }
    }
    vector<int> que;
    for (int i = 0; i < n; i++) {
      if (deg[i] == 0) {
        que.push_back(i);
      }
    }
    for (int b = 0; b < (int) que.size(); b++) {
      for (int eid : g[que[b]]) {
        auto& e = edges[eid];
        if (e.c - e.f > eps) {
          deg[e.to] -= 1;
          if (deg[e.to] == 0) {
            que.push_back(e.to);
          }
        }
      }
    }
    fill(pot.begin(), pot.end(), INF_C);
    pot[st] = 0;
    if (static_cast<int>(que.size()) == n) {
      for (int v : que) {
        if (pot[v] < INF_C) {
          for (int eid : g[v]) {
            auto& e = edges[eid];
            if (e.c - e.f > eps) {
              if (pot[v] + e.cost < pot[e.to]) {
                pot[e.to] = pot[v] + e.cost;
                pe[e.to] = eid;
              }
            }
          }
        }
      }
    } else {
      que.assign(1, st);
      vector<bool> in_queue(n, false);
      in_queue[st] = true;
      for (int b = 0; b < (int) que.size(); b++) {
        int i = que[b];
        in_queue[i] = false;
        for (int id : g[i]) {
          const edge &e = edges[id];
          if (e.c - e.f > eps && pot[i] + e.cost <
              pot[e.to]) {
            pot[e.to] = pot[i] + e.cost;
            pe[e.to] = id;
            if (!in_queue[e.to]) {
              que.push_back(e.to);
              in_queue[e.to] = true;
            }
          }
        }
      }
    }
    while (pot[fin] < INF_C) {
      T push = numeric_limits<T>::max();
      int v = fin;
      while (v != st) {
        const edge &e = edges[pe[v]];
        push = min(push, e.c - e.f);
        v = e.from;
      }
      v = fin;
      while (v != st) {
        edge &e = edges[pe[v]];
        e.f += push;
        edge &back = edges[pe[v] ^ 1];
        back.f -= push;
        v = e.from;
      }
      flow += push;
      cost += push * pot[fin];
      expath(st);
    }
    return {flow, cost};
  }
};
```

4.13 SCC

```
struct DirectedDfs {
  vector<vector<int>> g;
  int n;
  vector<int> num, low, current, S;
  int cnt;
  vector<int> comp_ids;
  vector<vector<int>> scc;

  DirectedDfs(const vector<vector<int>>& _g)
      : g(_g),
        n(g.size()),
        num(n, -1),
        low(n, 0),
        current(n, 0),
        cnt(0),
        comp_ids(n, -1) {
    for (int i = 0; i < n; i++) {
      if (num[i] == -1) dfs(i);
    }
  }

  void dfs(int u) {
    low[u] = num[u] = cnt++;
    S.push_back(u);
    current[u] = 1;
    for (auto v : g[u]) {
      if (num[v] == -1) dfs(v);
      if (current[v]) low[u] = min(low[u], low[v]);
    }
    if (low[u] == num[u]) {
      scc.push_back(vector<int>());
      while (1) {
        int v = S.back();
        S.pop_back();
        current[v] = 0;
        scc.back().push_back(v);
        comp_ids[v] = ((int)scc.size()) - 1;
        if (u == v) break;
      }
    }
  }

  // build DAG of strongly connected components
  // Returns: adjacency list of DAG
  vector<vector<int>> build_scc_dag() {
    vector<vector<int>> dag(scc.size());
    for (int u = 0; u < n; u++) {
      int x = comp_ids[u];
      for (int v : g[u]) {
        int y = comp_ids[v];
        if (x != y) {
          dag[x].push_back(y);
        }
      }
    }
    return dag;
  }
};
```

4.14 spfa

```
#include<bits/stdc++.h>
typedef pair<int, int> ii;
const int MaxN = 1e5 + 5;
const int Inf = 1e9;
vector<vector<ii>> AdjList;
int Dist[MaxN];
int Cnt[MaxN];
bool inqueue[MaxN];
int S;
int N;
queue<int> q;
```

```
bool spfa() {
    for(int i = 1 ; i <= N ; i++) {
        Dist[i] = Inf;
        Cnt[i] = 0;
        inqueue[i] = false;
    }
    Dist[S] = 0;
    q.push(S);
    inqueue[S] = true;
    while(!q.empty()) {
        int u = q.front();
        q.pop();
        inqueue[u] = false;

        for (ii tmp: AdjList[u]) {
            int v = tmp.first;
            int w = tmp.second;

            if (Dist[u] + w < Dist[v]) {
                Dist[v] = Dist[u] + w;
                if (!inqueue[v]) {
                    q.push(v);
                    inqueue[v] = true;
                    Cnt[v]++;
                    if (Cnt[v] > N)
                        return false;
                }
            }
        }
    }
    return true;
}
```

4.15 VirtualTree

```
bool anc(int u, int v) { return in[u] <= in[v] && in[v] <=
     out[u]; }
int solve(vi& ver) {
       for (int& i : ver) matter[i] = 1;
       all = (int)ver.size();
       sort(ver.begin(), ver.end(),
               [&](const int& a, const int& b) { return
                    in[a] < in[b]; });
       for (int i = 1; i < all; i++) {
              ver.pb(lca(ver[i - 1], ver[i]));
       }
       sort(ver.begin(), ver.end(),
               [&](const int& a, const int& b) { return
                    in[a] < in[b]; });
       ver.erase(unique(ver.begin(), ver.end()),
            ver.end());
       vi stk;
       stk.pb(ver[0]);
       for (int i = 1; i < (int)ver.size(); i++) {
              int u = ver[i];
              while (!stk.empty() && !anc(stk.back(), u))
                   stk.pop_back();
              g[stk.back()].pb(edges.size());
              g[ver[i]].pb(edges.size());
              edges.push_back({stk.back(), ver[i],
                   dis(stk.back(), ver[i])});
              stk.pb(ver[i]);
       }
       tot = 0;
       dfs(ver[0]);
       edges.clear();
       for (int i : ver) {
              matter[i] = 0;
              g[i].clear();
       }
       return tot;
}
```

5 Math

5.1 CRT

```
template<typename T>
T extgcd(T a, T b, T &x, T &y) {
  if (a == 0) {
    x = 0;
    y = 1;
    return b;
  }
  T p = b / a;
  T g = extgcd(b - p * a, a, y, x);
  x -= p * y;
  return g;
}

template<typename T>
bool diophantine(T a, T b, T c, T &x, T &y, T &g) {
  if (a == 0 && b == 0) {
    if (c == 0) {
      x = y = g = 0;
      return true;
    }
    return false;
  }
  if (a == 0) {
    if (c % b == 0) {
      x = 0;
      y = c / b;
      g = abs(b);
      return true;
    }
    return false;
  }
  if (b == 0) {
    if (c % a == 0) {
      x = c / a;
      y = 0;
      g = abs(a);
      return true;
    }
    return false;
  }
  g = extgcd(a, b, x, y);
  if (c % g != 0) {
    return false;
  }
  T dx = c / a;
  c -= dx * a;
  T dy = c / b;
  c -= dy * b;
  x = dx + (T) ((__int128) x * (c / g) % b);
  y = dy + (T) ((__int128) y * (c / g) % a);
  g = abs(g);
  return true;
  // |x|, |y| <= max(|a|, |b|, |c|) [tested]
}

bool crt(int64_t k1, int64_t m1, int64_t k2, int64_t m2,
     int64_t &k, int64_t &m) {
  k1 %= m1;
  if (k1 < 0) k1 += m1;
  k2 %= m2;
  if (k2 < 0) k2 += m2;
  int64_t x, y, g;
  if (!diophantine(m1, -m2, k2 - k1, x, y, g)) {
    return false;
  }
  int64_t dx = m2 / g;
  int64_t delta = x / dx - (x % dx < 0);
  k = m1 * (x - dx * delta) + k1;
  m = m1 / g * m2;
  assert(0 <= k && k < m);
  return true;
}

// for distinct prime modulos
template <typename T>
void crt_garner(const vector<int>& p, const vector<int>&
     a, T& res) {
  assert(p.size() == a.size());
  auto inverse = [&](int q, int m) {
    q %= m;
    if (q < 0) q += m;
    int b = m, u = 0, v = 1;
    while (q) {
      int t = b / q;
      b -= t * q; swap(q, b);
      u -= t * v; swap(u, v);
    }
    assert(b == 1);
    if (u < 0) u += m;
    return u;
  };
  vector<int> x(p.size());
  for (int i = 0; i < (int) p.size(); i++) {
    assert(0 <= a[i] && a[i] < p[i]);
    x[i] = a[i];
    for (int j = 0; j < i; j++) {
      x[i] = int(int64_t(x[i] - x[j]) * inverse(p[j], p[i])
           % p[i]);
      if (x[i] < 0) x[i] += p[i];
    }
  }
```

```
  res = 0;
  for (int i = (int) p.size() - 1; i >= 0; i--) {
    res = res * p[i] + x[i];
  }
}
```

5.2 Euclid

```
// x, y such that ax + by = gcd(a, b)
ll gcd(ll a, ll b) { return __gcd(a, b); }
ll euclid(ll a, ll b, ll &x, ll &y) {
  if (b) {
    ll d = euclid(b, a % b, y, x);
    return y -= a / b * x, d;
  }
  return x = 1, y = 0, a;
}
```

5.3 Factorization

```
inline long long qpow(long long a, int b) {
  long long ans = 1;
  while (b) {
    if (b & 1) ans = ans * a % mod;
    a = a * a % mod;
    b >>= 1;
  }
  return ans;
}
inline long long rv(int x) { return qpow(x, mod - 2) %
     mod; }
bool is_prime(long long n) {
  if (n <= 1) return false;
  for (int a : {2, 3, 5, 13, 19, 73, 193, 407521,
       299210837}) {
    if (n == a) return true;
    if (n % a == 0) return false;
  }
  long long d = n - 1;
  while (!(d & 1)) d >>= 1;
  for (int a : {2, 325, 9375, 28178, 450775, 9780504,
       1795265022}) {
    long long t = d, y = ipow(a, t, n);
    while (t != n - 1 && y != 1 && y != n - 1) y = mul(y,
         y, n), t <<= 1;
    if (y != n - 1 && !(t & 1)) return false;
  }
  return true;
}

long long pollard(long n) {
  auto f = [n](long x) { return mul(x, x, n) + 1; };
  long long x = 0, y = 0, t = 0, prd = 2, i = 1, q;
  while (t++ % 40 || gcd(prd, n) == 1) {
    if (x == y) x = ++i, y = f(x);
    if ((q = mul(prd, max(x, y) - min(x, y), n))) prd = q;
    x = f(x), y = f(f(y));
  }
  return gcd(prd, n);
}
vector<long long> factor(long n) {
  if (n == 1) return {};
  if (is_prime(n)) return {n};
  long x = pollard(n);
  auto l = factor(x), r = factor(n / x);
  l.insert(l.end(), r.begin(), r.end());
  return l;
}
```

5.4 FastSubsetTransform

```
// fast and/or/xor convolution
void FST(vi& a, bool inv) {
  for (int n = sz(a), step = 1; step < n; step *= 2) {
    for (int i = 0; i < n; i += 2 * step) rep(j, i, i +
         step) {
      int &u = a[j], &v = a[j + step];
      tie(u, v) = inv ? pii(v - u, u) : pii(v, u + v); //
           AND
      inv ? pii(v, u - v) : pii(u + v, u);          // OR
      pii(u + v, u - v);                            // XOR
    }
  }
  if (inv)
    for (int& x : a) x /= sz(a); // XOR only
}
vi conv(vi a, vi b) {
  FST(a, 0);
  FST(b, 0);
  rep(i, 0, sz(a)) a[i] *= b[i];
  FST(a, 1);
  return a;
}
```

5.5 FFT

```
// FFT / convolution. O(n log n). Exact for results up to
     ~1e14 (double).
// conv(a, b)[k] = sum_{i+j=k} a[i]*b[j]
typedef complex<double> cd;
void fft(vector<cd>& a, bool inv = false) {
      int n = sz(a), L = __builtin_ctz(n);
      static vector<cd> rt(2, 1);
      for (static int k = 2; k < n; k *= 2) {
            rt.resize(2 * k);
            long double ang = acosl(-1) / k;
            rep(j, 0, k) rt[k + j] = cd(cosl(ang * j),
                 sinl(ang * j));
      }
      vi rev(n);
      rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) /
           2;
      rep(i, 0, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
      for (int k = 1; k < n; k *= 2)
            for (int i = 0; i < n; i += 2 * k) rep(j, 0,
                 k) {
                  cd z = rt[j + k] * a[i + j + k];
                  a[i + j + k] = a[i + j] - z;
                  a[i + j] += z;
            }
      if (inv) {
            reverse(a.begin() + 1, a.end());
            for (cd& x : a) x /= n;
      }
}
vl conv(const vl& a, const vl& b) {
      if (a.empty() || b.empty()) return {};
      int m = sz(a) + sz(b) - 1, n = 1;
      while (n < m) n *= 2;
      vector<cd> f(n);
      rep(i, 0, sz(a)) f[i].real(a[i]);
      rep(i, 0, sz(b)) f[i].imag(b[i]);
      fft(f);
      for (cd& x : f) x *= x;         // (A+iB)^2: Im =
           2AB
      fft(f, true);
      vl res(m);
      rep(i, 0, m) res[i] = llround(f[i].imag() / 2);
      return res;
}
```

5.6 FFTmod

```
/**
 * Author: Ludo Pulles, chilli, Simon Lindholm
 * Date: 2019-01-09
 * License: CC0
 * Source:
      http://neerc.ifmo.ru/trains/toulouse/2017/fft2.pdf
      (do read, it's
 excellent) Accuracy bound from
      http://www.daemonology.net/papers/fft.pdf
 * Description: fft(a) computes $\hat f(k) = \sum_x a[x]
      \exp(2\pi i \cdot k x /
 N)$ for all $k$. N must be a power of 2. Useful for
      convolution:
   \texttt{conv(a, b) = c}, where $c[x] = \sum a[i]b[x-i]$.
   For convolution of complex numbers or more than two
        vectors: FFT, multiply
   pointwise, divide by n, reverse(start+1, end), FFT back.
   Rounding is safe if $(\sum a_i^2 + \sum b_i^2)\log_2{N}
        < 9\cdot10^{14}$
   (in practice $10^{16}$; higher for random inputs).
   Otherwise, use NTT/FFTMod.
 * Time: O(N \log N) with $N = |A|+|B|$ ($\tilde 1s$ for
      $N=2^{22}$)
 * Status: somewhat tested
 * Details: An in-depth examination of precision for both
      FFT and FFTMod can be
 found
 * here
```

```
 (https://github.com/simonlindholm/fft-precision/blob/master/fft-precision.md)
 */

typedef complex<double> C;
void fft(vector<C> &a) {
	int n = sz(a), L = 31 - __builtin_clz(n);
	static vector<complex<long double>> R(2, 1);
	static vector<C> rt(2, 1); // (^ 10% faster if
	    double)
	for (static int k = 2; k < n; k *= 2) {
		R.resize(n);
		rt.resize(n);
		auto x = polar(1.0L, acos(-1.0L) / k);
		rep(i, k, 2 * k) rt[i] = R[i] = i & 1 ? R[i
		    / 2] * x : R[i / 2];
	}
	vi rev(n);
	rep(i, 0, n) rev[i] = (rev[i / 2] | (i & 1) << L) /
	    2;
	rep(i, 0, n) if (i < rev[i]) swap(a[i], a[rev[i]]);
	for (int k = 1; k < n; k *= 2)
		for (int i = 0; i < n; i += 2 * k) rep(j, 0,
		    k) {
			// C z = rt[j+k] * a[i+j+k];
			    // (25% faster if
			    hand-rolled) ///
			// include-line
			auto x = (double *)&rt[j + k],
			    y = (double *)&a[i + j +
			    k]; /// exclude-line
			C z(x[0] * y[0] - x[1] * y[1],
			    x[0] * y[1] + x[1] *
			    y[0]); ///
			    exclude-line
			a[i + j + k] = a[i + j] - z;
			a[i + j] += z;
		}
}

typedef vector<ll> vl;
template <int M> vl convMod(const vl &a, const vl &b) {
	if (a.empty() || b.empty()) return {};
	vl res(sz(a) + sz(b) - 1);
	int B = 32 - __builtin_clz(sz(res)), n = 1 << B,
	    cut = int(sqrt(M));
	vector<C> L(n), R(n), outs(n), outl(n);
	rep(i, 0, sz(a)) L[i] = C((int)a[i] / cut,
	    (int)a[i] % cut);
	rep(i, 0, sz(b)) R[i] = C((int)b[i] / cut,
	    (int)b[i] % cut);
	fft(L), fft(R);
	rep(i, 0, n) {
		int j = -i & (n - 1);
		outl[j] = (L[i] + conj(L[j])) * R[i] / (2.0
		    * n);
		outs[j] = (L[i] - conj(L[j])) * R[i] / (2.0
		    * n) / 1i;
	}
	fft(outl), fft(outs);
	rep(i, 0, sz(res)) {
		ll av = ll(real(outl[i]) + .5), cv =
		    ll(imag(outs[i]) + .5);
		ll bv = ll(imag(outl[i]) + .5) +
		    ll(real(outs[i]) + .5);
		res[i] = ((av % M * cut + bv) % M * cut +
		    cv) % M;
	}
	return res;
}
```

5.7 Interpolate

```
const int mod = 1e9 + 7;
const int N = 1e6 + 6;

long long inv[N], po[N], pre[N], suf[N], dakdak[N];
long long ans, num;

inline long long qpow(long long a, int b) {
  long long ans = 1;
  while (b) {
    if (b & 1) ans = ans * a % mod;
    a = a * a % mod;
    b >>= 1;
  }
  return ans;
}
inline long long rv(int x) { return qpow(x, mod - 2) %
    mod; }
void prec() {
  inv[0] = 1;
  for (int i = 1; i <= k + 1; ++i) {
    inv[i] = (1LL * inv[i - 1] * rv(i)) % mod;
    po[i] = (po[i - 1] + qpow(i, k)) % mod;
  }
  for (int i = 1; i <= k + 1; ++i) {
    dakdak[i] = (inv[i] * inv[k + 1 - i]) % mod;
  }
}
inline long long interpolate(int x, int k, bool bf =
    false) {
  if (k == 0) return x;
  if (x <= k + 1 || bf) {
    return po[x];
  }
  pre[0] = x;
  suf[k + 1] = x - (k + 1);
  for (int i = 1; i <= k; i++) pre[i] = (pre[i - 1] * (x -
      i)) % mod;
  for (int i = k; i >= 1; i--) suf[i] = (suf[i + 1] * (x -
      i)) % mod;
  ans = 0;
  for (int i = 0; i <= k + 1; i++) {
    if (i == 0)
      num = suf[1];
    else if (i == k + 1)
      num = pre[k];
    else
      num = (pre[i - 1] * suf[i + 1]) % mod; // numerator

    if ((i + k) & 1)
      ans = (ans + ((po[i] * num % mod) * dakdak[i])) % mod;
    else
      ans = (ans - ((po[i] * num % mod) * dakdak[i])) % mod;

    ans = (ans + mod) % mod;
  }
  return ans;
}
```

5.8 Lucas

```
// computing nCk mod p in O(p + log_p n)
// if p isn't prime -> prime factorize p = p1^a1 * p2^a2 *
    ... * pr^ar
// -> calculate nCk mod p1, mod p2, ..., mod pr
// -> use chinese remainder theorem to recover nCk mod p
ll lucas(ll n, ll k, int p, vi& fact, vi& invfact) {
  ll c = 1;
  while (n || k) {
    ll a = n % p, b = k % p;
    if (a < b) return 0;
    c = c * fact[a] % p * invfact[b] % p * invfact[a - b] %
        p;
    n /= p;
    k /= p;
  }
  return c;
}
```

5.9 Matrix

```
template <class T, int N>
struct Matrix {
  typedef Matrix M;
  array<array<T, N>, N> d{};
  M operator*(const M& m) const {
    M a;
    rep(i, 0, N) rep(j, 0, N) rep(k, 0, N) a.d[i][j] +=
        d[i][k] * m.d[k][j];
    return a;
  }
  vector<T> operator*(const vector<T>& vec) const {
    vector<T> ret(N);
    rep(i, 0, N) rep(j, 0, N) ret[i] += d[i][j] * vec[j];
    return ret;
  }
  M operator(ll p) const {
    assert(p >= 0);
    M a, b(*this);
    rep(i, 0, N) a.d[i][i] = 1;
    while (p) {
      if (p & 1) a = a * b;
      b = b * b;
      p >>= 1;
    }
```

```
    return a;
  }
};
```

5.10 MillerRabin

```
inline uint64_t mod_mult64(uint64_t a, uint64_t b,
     uint64_t m) { return __int128_t(a) * b % m; }
uint64_t mod_pow64(uint64_t a, uint64_t b, uint64_t m) {
  uint64_t ret = (m > 1);
  for (;;) {
    if (b & 1) ret = mod_mult64(ret, a, m);
    if (!(b >>= 1)) return ret;
    a = mod_mult64(a, a, m);
  }
}

// Works for all primes p < 2^64
bool is_prime(uint64_t n) {
  if (n <= 3) return (n >= 2);
  static const uint64_t small[] = {
      2,  3,  5,   7,   11,  13,  17,  19,  23,  29,  31,  37,
           41,  43,  47,  53,  59,  61,  67,  71,  73,  79,  83,
      89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139,
           149, 151, 157, 163, 167, 173, 179, 181, 191,
           193, 197, 199,
  };
  for (size_t i = 0; i < sizeof(small) / sizeof(uint64_t);
       ++i) {
    if (n % small[i] == 0) return n == small[i];
  }

  // Makes use of the known bounds for Miller-Rabin
       pseudoprimes.
  static const uint64_t millerrabin[] = {
      2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37,
  };
  static const uint64_t A014233[] = {
      // From OEIS.
      2047LL, 1373653LL, 25326001LL, 3215031751LL,
           2152302898747LL, 3474749660383LL,
           341550071728321LL, 341550071728321LL,
           3825123056546413051LL, 3825123056546413051LL,
           3825123056546413051LL, 0,
  };
  uint64_t s = n - 1, r = 0;
  while (s % 2 == 0) {
    s /= 2;
    r++;
  }
  for (size_t i = 0, j; i < sizeof(millerrabin) /
       sizeof(uint64_t); i++) {
    uint64_t md = mod_pow64(millerrabin[i], s, n);
    if (md != 1) {
      for (j = 1; j < r; j++) {
        if (md == n - 1) break;
        md = mod_mult64(md, md, n);
      }
      if (md != n - 1) return false;
    }
    if (n < A014233[i]) return true;
  }
  return true;
}
```

5.11 Mobius

```
mobius[1] = 1;
for (int i = 2; i < N; ++i) {
  --mobius[i];
  for (int j = i + i; j < N; j += i) mobius[j] -=
       mobius[i];
}
```

5.12 ModInverse

```
const ll mod = 1000000007, LIM = 200000;
ll* inv = new ll[LIM] - 1; inv[1] = 1;
for(ll i = 2; i < LIM; ++i) inv[i] = mod - (mod / i) *
     inv[mod % i] % mod;
```

5.13 ModMulLL

```
typedef unsigned long long ull;
const int bits = 10; // i f a l l numbers are less than
     2^k , set bits = 64k
const ull po = 1 << bits;
ull mod_mul(ull a, ull b, ull &c) {
  ull x = a * (b & (po - 1)) % c;
  while ((b >>= bits) > 0) {
    a = (a << bits) % c;
    x += (a * (b & (po - 1))) % c;
  }
  return x % c;
}
ull mod_pow(ull a, ull b, ull mod) {
  if (b == 0) return 1;
  ull res = mod_pow(a, b / 2, mod);
  res = mod_mul(res, res, mod);
  if (b & 1) return mod_mul(res, a, mod);
  return res;
}
```

5.14 ModularArithmetic

```
const ll mod = 17; // change to something else
struct Mod {
  ll x;
  Mod(ll xx) : x(xx) {}
  Mod operator+(Mod b) { return Mod((x + b.x) % mod); }
  Mod operator-(Mod b) { return Mod((x - b.x + mod) %
       mod); }
  Mod operator*(Mod b) { return Mod((x * b.x) % mod); }
  Mod operator/(Mod b) { return *this * invert(b); }
  Mod invert(Mod a) {
    ll x, y, g = euclid(a.x, mod, x, y);
    assert(g == 1);
    return Mod((x + mod) % mod);
  }
  Mod operator^(ll e) {
    if (!e) return Mod(1);
    Mod r = *this ^ (e / 2);
    r = r * r;
    return e & 1 ? *this * r : r;
  }
};
```

5.15 Notes

5.15.1 Cycles

Let $g_S(n)$ be the number of n-permutations whose cycle lengths all belong to the set S. Then

$$\sum_{n=0}^{\infty} g_S(n)\frac{x^n}{n!} = \exp\left(\sum_{n\in S}\frac{x^n}{n}\right)$$

5.15.2 Derangements

Permutations of a set such that none of the elements appear in their original position.

$$D(n) = (n-1)(D(n-1)+D(n-2)) = nD(n-1)+(-1)^n = \left\lfloor \frac{n!}{e} \right\rceil$$

5.15.3 Burnside's lemma

Given a group G of symmetries and a set X, the number of elements of X *up to symmetry* equals

$$\frac{1}{|G|}\sum_{g\in G}|X^g|,$$

where X^g are the elements fixed by g ($g.x = x$).

If $f(n)$ counts "configurations" (of some sort) of length n, we can ignore rotational symmetry using $G = Z_n$ to get

$$g(n) = \frac{1}{n}\sum_{k=0}^{n-1} f(\gcd(n,k)) = \frac{1}{n}\sum_{k|n} f(k)\phi(n/k).$$

5.15.4 Partition function

Number of ways of writing n as a sum of positive integers, disregarding the order of the summands.

$$p(0) = 1,\ p(n) = \sum_{k \in Z \setminus \{0\}} (-1)^{k+1} p(n - k(3k-1)/2)$$

$$p(n) \sim 0.145/n \cdot \exp(2.56\sqrt{n})$$

n	0	1	2	3	4	5	6	7	8	9	20	50	100
$p(n)$	1	1	2	3	5	7	11	15	22	30	627	∼2e5	∼2e8

5.15.5 Lucas' Theorem

Let n, m be non-negative integers and p a prime. Write $n = n_k p^k + ... + n_1 p + n_0$ and $m = m_k p^k + ... + m_1 p + m_0$. Then $\binom{n}{m} \equiv \prod_{i=0}^{k} \binom{n_i}{m_i} \pmod{p}$.

5.15.6 Bernoulli numbers

EGF of Bernoulli numbers is $B(t) = \frac{t}{e^t - 1}$ (FFT-able). $B[0, \ldots] = [1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42}, \ldots]$

Sums of powers:

$$\sum_{i=1}^{n} n^m = \frac{1}{m+1} \sum_{k=0}^{m} \binom{m+1}{k} B_k \cdot (n+1)^{m+1-k}$$

Euler-Maclaurin formula for infinite sums:

$$\sum_{i=m}^{\infty} f(i) = \int_m^{\infty} f(x)dx - \sum_{k=1}^{\infty} \frac{B_k}{k!} f^{(k-1)}(m)$$

$$\approx \int_m^{\infty} f(x)dx + \frac{f(m)}{2} - \frac{f'(m)}{12} + \frac{f'''(m)}{720} + O(f^{(5)}(m))$$

5.15.7 Stirling numbers of the first kind

Number of permutations on n items with k cycles.

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k),\ c(0,0) = 1$$

$$\sum_{k=0}^{n} c(n,k)x^k = x(x+1)\ldots(x+n-1)$$

$c(8,k) = 8, 0, 5040, 13068, 13132, 6769, 1960, 322, 28, 1$
$c(n,2) = 0, 0, 1, 3, 11, 50, 274, 1764, 13068, 109584, \ldots$

5.15.8 Eulerian numbers

Number of permutations $\pi \in S_n$ in which exactly k elements are greater than the previous element. k j:s s.t. $\pi(j) > \pi(j+1)$, $k+1$ j:s s.t. $\pi(j) \geq j$, k j:s s.t. $\pi(j) > j$.

$$E(n,k) = (n-k)E(n-1,k-1) + (k+1)E(n-1,k)$$

$$E(n,0) = E(n,n-1) = 1$$

$$E(n,k) = \sum_{j=0}^{k} (-1)^j \binom{n+1}{j} (k+1-j)^n$$

5.15.9 Stirling numbers of the second kind

Partitions of n distinct elements into exactly k groups.

$$S(n,k) = S(n-1,k-1) + kS(n-1,k)$$

$$S(n,1) = S(n,n) = 1$$

$$S(n,k) = \frac{1}{k!} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} j^n$$

5.15.10 Bell numbers

Total number of partitions of n distinct elements. $B(n) = 1, 1, 2, 5, 15, 52, 203, 877, 4140, 21147, \ldots.$ For p prime,

$$B(p^m + n) \equiv mB(n) + B(n+1) \pmod{p}$$

5.15.11 Labeled unrooted trees

on n vertices: n^{n-2}
on k existing trees of size n_i: $n_1 n_2 \cdots n_k n^{k-2}$
with degrees d_i: $(n-2)!/((d_1-1)! \cdots (d_n-1)!)$

5.15.12 Catalan numbers

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1} = \frac{(2n)!}{(n+1)!n!}$$

$$C_0 = 1,\ C_{n+1} = \frac{2(2n+1)}{n+2} C_n,\ C_{n+1} = \sum C_i C_{n-i}$$

$C_n = 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, \ldots$

- sub-diagonal monotone paths in an $n \times n$ grid.
- strings with n pairs of parenthesis, correctly nested.
- binary trees with with $n+1$ leaves (0 or 2 children).
- ordered trees with $n+1$ vertices.
- ways a convex polygon with $n+2$ sides can be cut into triangles by connecting vertices with straight lines.
- permutations of $[n]$ with no 3-term increasing subseq.

5.15.13 Hockey Stick Identity

$$\sum_{k=r}^{n} \binom{k}{r} = \binom{n+1}{r+1}$$

5.16 NTT

```
const int mod = 998244353;

ll power(ll a, ll b) {
	ll res = 1;
	while (b) {
		if (b & 1) res = res * a % mod;
		a = a * a % mod; b >>= 1;
	}
	return res;
}

void ntt(vll &a, bool invert = false) {
	int n = a.size();
	int LG = __builtin_ctz(n);
	vi rev(n);
	for (int i = 1; i < n; i++)
		rev[i] = rev[i >> 1] >> 1 | (i & 1) << (LG -
		    1);
	for (int i = 0; i < n; i++) {
		if (i < rev[i]) {
			swap(a[i], a[rev[i]]);
		}
	}
	int base = power(3, (mod - 1) / n);
	for (int len = 2; len <= n; len <<= 1) {
		int wlen = invert ? power(base, n - n / len)
		    : power(base, n / len);
		for (int i = 0; i < n; i += len) {
			ll w = 1;
			for (int j = 0; j < len / 2; j++) {
				ll u = a[i + j], v = a[i + j
				    + len / 2] * w % mod;
				a[i + j] = (u + v < mod ? u +
				    v : u + v - mod);
				a[i + j + len / 2] = (u >= v
				    ? u - v : u - v + mod);
				(w *= wlen) %= mod;
			}
		}
	}
	if (invert) {
		ll divn = power(n, mod - 2);
		for (auto &x : a) {
			(x *= divn) %= mod;
		}
	}
}

vll convolution(vll a, vll b) {
	int n = 1;
	while (n < a.size() + b.size()) n <<= 1;
	a.resize(n);
	b.resize(n);
	ntt(a);
	ntt(b);
	for (int i = 0; i < n; i++) {
		a[i] *= b[i];
		a[i] %= mod;
	}
```

```
    ntt(a, true);
    return a;
}
```

5.17 PhiFunction

```
const int LIM = 5000000;
int phi[LIM];
void calculatePhi() {
  rep(i, 0, LIM) phi[i] = i & 1 ? i : i / 2;
  for (int i = 3; i < LIM; i += 2)
    if (phi[i] == i)
      for (int j = i; j < LIM; j += i) (phi[j] /= i) *= i -
          1;
```

5.18 PollardFactorize

```
using ll = long long;
using ull = unsigned long long;
using ld = long double;
ll mult(ll x, ll y, ll md) {
  ull q = (ld)x * y / md;
  ll res = ((ull)x * y - q * md);
  if (res >= md) res -= md;
  if (res < 0) res += md;
  return res;
}

ll powMod(ll x, ll p, ll md) {
  if (p == 0) return 1;
  if (p & 1) return mult(x, powMod(x, p - 1, md), md);
  return powMod(mult(x, x, md), p / 2, md);
}

bool checkMillerRabin(ll x, ll md, ll s, int k) {
  x = powMod(x, s, md);
  if (x == 1) return true;
  while (k--) {
    if (x == md - 1) return true;
    x = mult(x, x, md);
    if (x == 1) return false;
  }
  return false;
}
bool isPrime(ll x) {
  if (x == 2 || x == 3 || x == 5 || x == 7) return true;
  if (x % 2 == 0 || x % 3 == 0 || x % 5 == 0 || x % 7 ==
      0) return false;
  if (x < 121) return x > 1;
  ll s = x - 1;
  int k = 0;
  while (s % 2 == 0) {
    s >>= 1;
    k++;
  }
  if (x < 1LL << 32) {
    for (ll z : {2, 7, 61}) {
      if (!checkMillerRabin(z, x, s, k)) return false;
    }
  } else {
    for (ll z : {2, 325, 9375, 28178, 450775, 9780504,
         1795265022}) {
      if (!checkMillerRabin(z, x, s, k)) return false;
    }
  }
  return true;
}

ll gcd(ll x, ll y) { return y == 0 ? x : gcd(y, x % y); }

void pollard(ll x, vector<ll> &ans) {
  if (isPrime(x)) {
    ans.push_back(x);
    return;
  }
  ll c = 1;
  while (true) {
    c = 1 + get_rand(x - 1);
    auto f = [&](ll y) {
      ll res = mult(y, y, x) + c;
      if (res >= x) res -= x;
      return res;
    };
    ll y = 2;
    int B = 100;
    int len = 1;
    ll g = 1;
    while (g == 1) {
      ll z = y;
      for (int i = 0; i < len; i++) {
        z = f(z);
      }
      ll zs = -1;
      int lft = len;
      while (g == 1 && lft > 0) {
        zs = z;
        ll p = 1;
        for (int i = 0; i < B && i < lft; i++) {
          p = mult(p, abs(z - y), x);
          z = f(z);
        }
        g = gcd(p, x);
        lft -= B;
      }
      if (g == 1) {
        y = z;
        len <<= 1;
        continue;
      }
      if (g == x) {
        g = 1;
        z = zs;
        while (g == 1) {
          g = gcd(abs(z - y), x);
          z = f(z);
        }
      }
      if (g == x) break;
      assert(g != 1);
      pollard(g, ans);
      pollard(x / g, ans);
      return;
    }
  }
}
// return list of all prime factors of x (can have
    duplicates)
vector<ll> factorize(ll x) {
  vector<ll> ans;
  for (ll p : {2, 3, 5, 7, 11, 13, 17, 19}) {
    while (x % p == 0) {
      x /= p;
      ans.push_back(p);
    }
  }
  if (x != 1) {
    pollard(x, ans);
  }
  sort(ans.begin(), ans.end());
  return ans;
}
// return pairs of (p, k) where x = product(p^k)
vector<pair<ll, int>> factorize_pk(ll x) {
  auto ps = factorize(x);
  ll last = -1, cnt = 0;
  vector<pair<ll, int>> res;
  for (auto p : ps) {
    if (p == last)
      ++cnt;
    else {
      if (last > 0) res.emplace_back(last, cnt);
      last = p;
      cnt = 1;
    }
  }
  if (cnt > 0) {
    res.emplace_back(last, cnt);
  }
  return res;
}
vector<ll> get_all_divisors(ll n) {
  auto pks = factorize_pk(n);

  vector<ll> res;
  function<void(int, ll)> gen = [&](int i, ll prod) {
    if (i == static_cast<int>(pks.size())) {
      res.push_back(prod);
      return;
    }

    ll cur_power = 1;
    for (int cur = 0; cur <= pks[i].second; ++cur) {
      gen(i + 1, prod * cur_power);
      cur_power *= pks[i].first;
    }
  };

  gen(0, 1LL);
  sort(res.begin(), res.end());
  return res;
```

```
}
```

5.19 PrimitiveRoot

```
// Primitive root of modulo n is integer g iff for all a <
     n & gcd(a, n) == 1, there exist k: g^k = a mod n
// k is called discrete log of a (in case P is prime, can
     find in O(sqrt(P)) by noting that (P-1) is divisible
     by k)
//
// Exist if:
// - n is 1, 2, 4
// - n = p^k for odd prime p
// - n = 2*p^k for odd prime p
int powmod (int a, int b, int p) {
    int res = 1;
    while (b)
        if (b & 1)
            res = int (res * 1ll * a % p), --b;
        else
            a = int (a * 1ll * a % p), b >>= 1;
    return res;
}

int generator (int p) {
    vector<int> fact;
    int phi = p-1, n = phi;
    for (int i=2; i*i<=n; ++i)
        if (n % i == 0) {
            fact.push_back (i);
            while (n % i == 0)
                n /= i;
        }
    if (n > 1)
        fact.push_back (n);

    for (int res=2; res<=p; ++res) {
        bool ok = true;
        for (size_t i=0; i<fact.size() && ok; ++i)
            ok &= powmod (res, phi / fact[i], p) != 1;
        if (ok) return res;
    }
    return -1;
}
```

5.20 TernarySearch

```
// Find the smallest i in [a; b] that maximizes f(i),
     assuming
// that f(a) < .. < f(i) >= ... >= f(b)
// Usage: int ind = ternSearch(0,n-1,[&](int i){return
     a[i];});
template <class F>
int ternSearch(int a, int b, F f) {
  assert(a <= b);
  while (b - a >= 5) {
    int mid = (a + b) / 2;
    if (f(mid) < f(mid + 1))
      a = mid; // (A)
    else
      b = mid + 1;
  }
  rep(i, a + 1, b + 1) if (f(a) < f(i)) a = i; // (B)
  return a;
}
```

5.21 XorBasis

```
struct XorBasis {
  vector<T> basis;
  bool add(T mask) {
    for (auto u : basis) mask = min(mask, mask ^ u);
    for (auto& u : basis) u = min(u, u ^ mask);
    if (mask) {
      basis.push_back(mask);
      return true;
    }
    return false;
  }
  T getMax() {
    T res = 0;
    for (auto u : basis) res = max(res, res ^ u);
    return res;
  }
  void init() { sort(basis.begin(), basis.end()); }
  bool InSpan(T mask) {
    for (auto u : basis) mask = min(mask, mask ^ u);
    return (mask == 0);
  }
  T getK(T k) // 0-indexed
  {
    T res = 0;
    for (int i = sz(basis) - 1; i >= 0; i--)
      if (k >> i & 1) res ^= basis[i];
    return res;
  }
};
```

6 Misc

6.1 HilbertOrder

```
inline int64_t hilbertOrder(int x, int y, int pow, int
     rotate) {
      if (pow == 0) {
            return 0;
      }
      int hpow = 1 << (pow-1);
      int seg = (x < hpow) ? (
            (y < hpow) ? 0 : 3
      ) : (
            (y < hpow) ? 1 : 2
      );
      seg = (seg + rotate) & 3;
      const int rotateDelta[4] = {3, 0, 0, 1};
      int nx = x & (x ^ hpow), ny = y & (y ^ hpow);
      int nrot = (rotate + rotateDelta[seg]) & 3;
      int64_t subSquareSize = int64_t(1) << (2*pow - 2);
      int64_t ans = seg * subSquareSize;
      int64_t add = hilbertOrder(nx, ny, pow-1, nrot);
      ans += (seg == 1 || seg == 2) ? add :
           (subSquareSize - add - 1);
      return ans;
}
```

7 String

7.1 AhoCorasick

```
template <int MAXC = 26>
struct AhoCorasick {
  vector<array<int, MAXC>> C;
  vector<int> F;
  vector<vector<int>> FG;
  vector<bool> E;

  int node() {
    int r = C.size();
    E.push_back(0);
    F.push_back(-1);
    C.emplace_back();
    fill(C.back().begin(), C.back().end(), -1);
    return r;
  }
  int ctrans(int n, int c) {
    if (C[n][c] == -1) C[n][c] = node();
    return C[n][c];
  }
  int ftrans(int n, int c) const {
    while (n && C[n][c] == -1) n = F[n];
    return C[n][c] != -1 ? C[n][c] : 0;
  }
  AhoCorasick(vector<vector<int>> P) {
    node();
    for (int i = 0; i < (int)P.size(); i++) {
      int n = 0;
      for (int c : P[i]) n = ctrans(n, c);
      E[n] = 1;
    }
    queue<int> Q;
    F[0] = 0;
    for (int c : C[0])
      if (c != -1) Q.push(c), F[c] = 0;
    while (!Q.empty()) {
      int n = Q.front();
      Q.pop();
      for (int c = 0; c < MAXC; ++c)
        if (C[n][c] != -1) {
          int f = F[n];
```

```
        while (f && C[f][c] == -1) f = F[f];
        F[C[n][c]] = C[f][c] != -1 ? C[f][c] : 0;
        Q.emplace(C[n][c]);
      }
    }
    FG.resize(F.size());
    for (int i = 1; i < (int)F.size(); i++) {
      FG[F[i]].push_back(i);
      if (E[i]) Q.push(i);
    }
    while (!Q.empty()) {
      int n = Q.front();
      Q.pop();
      for (int f : FG[n]) E[f] = 1, Q.push(f);
    }
  }
  bool check(vector<int> V) {
    if (E[0]) return 1;
    int n = 0;
    for (int c : V) {
      n = ftrans(n, c);
      if (E[n]) return 1;
    }
    return 0;
  }
};
```

7.2 KMP

```
// prefix function: *length* of longest prefix which is
     also suffix:
//   pi[i] = max(k: s[0..k-1] == s[i-(k-1)..i]
//
// KMP {{{
template <typename Container>
std::vector<int> prefix_function(const Container& s) {
  int n = s.size();
  std::vector<int> pi(n);
  for (int i = 1; i < n; ++i) {
    int j = pi[i - 1];
    while (j > 0 && s[i] != s[j]) j = pi[j - 1];
    if (s[i] == s[j]) ++j;
    pi[i] = j;
  }
  return pi;
}

// Tested: https://oj.vnoi.info/problem/substr
// Return all positions (0-based) that pattern `pat`
     appears in `text`
std::vector<int> kmp(const std::string& pat, const
     std::string& text) {
  auto pi = prefix_function(pat + '\0' + text);
  std::vector<int> res;
  for (size_t i = pi.size() - text.size(); i < pi.size();
       ++i) {
    if (pi[i] == (int)pat.size()) {
      res.push_back(i - 2 * pat.size());
    }
  }
  return res;
}

// Tested: https://oj.vnoi.info/problem/icpc22_mt_b
// Returns cnt[i] = # occurrences of prefix of length-i
// NOTE: cnt[0] = n+1 (0-length prefix appears n+1 times)
std::vector<int> prefix_occurrences(const string& s) {
  int n = s.size();
  auto pi = prefix_function(s);
  std::vector<int> res(n + 1);
  for (int i = 0; i < n; ++i) res[pi[i]]++;
  for (int i = n - 1; i > 0; --i) res[pi[i - 1]] += res[i];
  for (int i = 0; i <= n; ++i) res[i]++;
  return res;
}
```

7.3 Manacher

```
vector<int> manacher_odd(string s) {
  int n = s.size();
  s = "$" + s + "^";
  vector<int> p(n + 2);
  int l = 1, r = 1;
  for (int i = 1; i <= n; i++) {
    p[i] = max(0, min(r - i, p[l + (r - i)]));
    while (s[i - p[i]] == s[i + p[i]]) {
      p[i]++;
    }
    if (i + p[i] > r) {
      l = i - p[i], r = i + p[i];
    }
  }
  return vector<int>(begin(p) + 1, end(p) - 1);
}
vector<int> manacher(string s) {
  string t;
  for (auto c : s) {
    t += string("#") + c;
  }
  auto res = manacher_odd(t + "#");
  return vector<int>(begin(res) + 1, end(res) - 1);
}
```

7.4 StringHashing

```
const int MOD1 = 127657753, MOD2 = 987654319;
const int p1 = 137, p2 = 277;
```

7.5 SuffixArray

```
/**
 * Author:        , chilli
 * Date: 2019-04-11
 * License: Unknown
 * Source: Suffix array - a powerful tool for dealing with
     strings
 * (Chinese IOI National team training paper, 2009)
 * Description: Builds suffix array for a string.
 * \texttt{sa[i]} is the starting index of the suffix which
 * is $i$'th in the sorted suffix array.
 * The returned vector is of size $n+1$, and \texttt{sa[0]
     = n}.
 * The \texttt{lcp} array contains longest common prefixes
     for
 * neighbouring strings in the suffix array:
 * \texttt{lcp[i] = lcp(sa[i], sa[i-1])}, \texttt{lcp[0] =
     0}.
 * The input string must not contain any zero bytes.
 * Time: O(n \log n)
 * Status: stress-tested
 */
struct SuffixArray {
       vi sa, lcp;
       SuffixArray(string& s, int lim=256) { // or
            basic_string<int>
              int n = sz(s) + 1, k = 0, a, b;
              vi x(all(s)), y(n), ws(max(n, lim));
              x.push_back(0), sa = lcp = y, iota(all(sa),
                   0);
              for (int j = 0, p = 0; p < n; j = max(1, j *
                   2), lim = p) {
                p = j, iota(all(y), n - j);
                rep(i,0,n) if (sa[i] >= j) y[p++] =
                     sa[i] - j;
                fill(all(ws), 0);
                rep(i,0,n) ws[x[i]]++;
                rep(i,1,lim) ws[i] += ws[i - 1];
                for (int i = n; i--;)
                     sa[--ws[x[y[i]]]] = y[i];
                swap(x, y), p = 1, x[sa[0]] = 0;
                rep(i,1,n) a = sa[i - 1], b = sa[i],
                     x[b] =
                       (y[a] == y[b] && y[a + j] ==
                            y[b + j]) ? p - 1 : p++;
              }
              for (int i = 0, j; i < n - 1; lcp[x[i++]] =
                   k)
                for (k && k--, j = sa[x[i] - 1];
                              s[i + k] == s[j + k];
                                   k++);
       }
};
int64_t cnt_distinct_substrings(const std::string& s) {
   auto lcp = LCP(s, suffix_array(s, 0, 255));
   return s.size() * (int64_t) (s.size() + 1) / 2
       - std::accumulate(lcp.begin(), lcp.end(), 0LL);
}
```

7.6 Z

```
vector<int> zfunc(const string& s) {
```

```
  int n = (int)s.length();
  vector<int> z(n);
  z[0] = n;
  for (int i = 1, l = 0, r = 0; i < n; ++i) {
    if (i <= r) z[i] = min(r - i + 1, z[i - l]);
    while (i + z[i] < n && s[z[i]] == s[i + z[i]]) ++z[i];
    if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
  }
  return z;
}
```

8 Utilities

8.1 compile

```
function cm() {
    echo "Compiling $1.cpp..."
    echo "g++ -std=c++20 -O2 $1.cpp -o $1"
    g++ -std=c++20 -O2 $1.cpp -o $1
}
```

8.2 debug

```
string to_string(string s) { return '"' + s + '"'; }
string to_string(const char* s) { return
     to_string((string)s); }
string to_string(bool b) { return (b ? "true" : "false"); }
template <typename A, typename B>
string to_string(pair<A, B> p) {
  return "(" + to_string(p.first) + ", " +
       to_string(p.second) + ")";
}
template <typename A>
string to_string(A v) {
  bool first = true;
  string res = "{";
  for (const auto& x : v) {
    if (!first) {
      res += ", ";
    }
    first = false;
    res += to_string(x);
  }
  res += "}";
  return res;
}
void debug_out() { cerr << endl; }
template <typename Head, typename... Tail>
void debug_out(Head H, Tail... T) {
  cerr << " " << to_string(H);
  debug_out(T...);
}
#ifdef LOCAL
#define debug(...) cerr << "[" << #__VA_ARGS__ << "]:",
     debug_out(__VA_ARGS__)
#else
#define debug(...) 42
#endif
```

8.3 FastInput

```
inline char gc() { // 1 ike getchar ()
  static char buf[1 << 16];
  static size_t bc, be;
  if (bc >= be) {
    buf[0] = 0, bc = 0;
    be = fread(buf, 1, sizeof(buf), stdin);
  }
  return buf[bc++]; // returns 0 on EOF
}
int readInt() {
  int a, c;
  while ((a = gc()) < 40)
    ;
  if (a == '-') return -readInt();
  while ((c = gc()) >= 48) a = a * 10 + c - 480;
  return a - 48;
}
```

8.4 pragma

```
#pragma GCC optimize("O3,unroll-loops")
#pragma GCC target("avx2,bmi,bmi2,popcnt,lzcnt")
```

8.5 template

```
#include <bits/stdc++.h>
using namespace std;
mt19937_64
     rng(chrono::steady_clock::now().time_since_epoch().count());
ll get_rand(ll l, ll r) {
    return uniform_int_distribution<ll> (l, r)(rng);
}
int32_t main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);
    cerr << "\n[Time]: " << 1000.0 * clock() /
         CLOCKS_PER_SEC << " ms.\n";
    return 0;
}
```

8.6 vimrc

```
source $VIMRUNTIME/defaults.vim
let mapleader = " "
runtime! ftplugin/man.vim
packadd! matchit
set modeline
set autoread
set hidden
set undofile
set undodir=~/.vim/undo//
set noswapfile
set tabstop=4
set softtabstop=4
set shiftwidth=4
set expandtab
set smarttab
set autoindent
set nojoinspaces
set nostartofline
set hlsearch
set ignorecase
set smartcase
set mouse=a
set background=dark
set number
set laststatus=2
set belloff=all
set splitbelow
set splitright
set scrolloff=2
nnoremap Y y$
nnoremap <C-L> :nohlsearch<Bar>normal! <C-L><CR>
nnoremap <Leader>h :tab h
nnoremap <Leader>e :e $MYVIMRC<CR>
nnoremap <Leader>r :source $MYVIMRC<CR>
nnoremap <Leader>c :w <Bar> !g++ -std=c++20 -O2 -Wall % -o
     %:r<CR>
nnoremap <Leader>x :!./%:r<CR>
```